

claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\workflows\wf_fc91ae6e-b62\agent-a69081c6ab1817fc4.jsonl`  
- SHA-256: `0dd3617f95331edae5851af846ec7a818919602ca1caeef05609f3d0b06eb8d5`  
- Source modified: `2026-06-14T06:57:43+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_fc91ae6e-b62`  
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`
```

Transcript

1. user (2026-06-14T06:52:32.048Z)

You are writing a PAPER-STYLE project quality report (for a professor; intended to graduate into a research paper) about a local, cheap, AI rally-detection system for racket sports. Below are VERIFIED facts/numbers gathered from the codebase ? use ONLY these numbers, accurately (do not invent or round differently):

```
[  
  {  
    "area": "Rally-detection quantitative results (badminton-highlight-indexer): baselines,  
Gate-A promotion, learned-variant fix #130, fusion P3/P4 ablations, golden corpus size",  
    "facts": [  
      "BASELINES TABLE (docs/QUALITY_ITERATIONS.md, 'Baselines that drive strategy', IoU 0.5 vs  
human golden labels). Two columns: Mahadevpura (clean-ish) and Kushagra (multi-court). Heuristic  
(velocity windowing): 0.657 / 0.157. Gen-0 owned head (LOVO, n=6): 0.744 / 0.561. Two-stage  
WASB+Gemini refine: 0.83 / (no Kushagra value, dash). Gemini wrapper (gemini-flash-latest): 0.93  
/ 0.66.",  
      "Doc reading note (verbatim): 'clean footage is commoditized (serve via Gemini); multi-  
court drops the commodity wrapper to 0.66 ? that is the owned model's ship target.' The owner-  
observed false positive (player holding/flicking shuttle between points counted as a rally) is  
the precision bug the fusion gates target.",  
      "GEMINI WRAPPER per-video detail (eval_baselines/gemini_wrapper_2026-06-10.json).  
mahadevpura_singles: f1 0.93, precision 1.00, recall 0.87, mean_iou 0.80, boundary_err_s 0.7,  
n_gt 31, n_pred 27 (3 of 4 FNs detected by Gemini but rejected by min_segment_duration=3.0  
filter on 2.5s rallies). kushagra_singles: f1 0.66, precision 0.69, recall 0.62, mean_iou 0.73,  
boundary_err_s 1.2, n_gt 32, n_pred 29 (multi-court, contrast 1.4x). Model resolved as gemini-  
flash-latest (3.5-flash era). The JSON's own comparison_at_time_of_run block restates  
heuristic_lovo {mahadevpura 0.657, kushagra 0.157} and gen0_learned_lovo {mahadevpura 0.744,  
kushagra 0.561}.",  
      "GEN-0 / HEURISTIC LOVO FLOOR (eval_baselines/heuristic_lovo_n6.json ?  
'heuristic_velocity_windowing_LOVO', the honest cross-video floor the Gen-0 learned model must  
beat; calibrated velocity-windowing heuristic, leave-one-video-out over n=6 golden corpus;  
computed 2026-06-10). mean_f1 0.480, std 0.242. Per-video F1: adarsh_avi_singles 0.751,  
kushagra_singles 0.157, largetest_doubles 0.737, gbaaddy 0.309, testlarge_short 0.267,  
mahadevpura_singles 0.657.",  
      "GEN-0 LEARNED LOVO SCORE + FLOOR + SIGN-TEST CONTEXT (docs/QUALITY_ITERATIONS.md archived  
predecessor '2026-06-multivideo-lovo'): at n=6, learned 0.583 vs heuristic 0.480 (learned per-  
frame prototype holds ~+0.10, overtook the heuristic at n=3); AUC 0.83-0.92 decoupled from  
contrast. Standing-lessons note on the sign test: 'Sign test needs n ? ~n=10/9-wins for p<0.05.'  
(i.e., n=6 cannot reach significance for that magnitude.)",  
      "GATE-A PROMOTION STATS (docs/QUALITY_ITERATIONS.md, 'Gate-A PROMOTED' entry, 2026-06-14;  
net-crossings >= 2, first cue promoted via the harness). Measured win: ?F1 +0.074, 6/6 golden  
matches, sign-test p=0.031, CI [+0.042, +0.104]; over-segmentation (segment-count ratio) 1.68 ->  
1.01. Recommended config: indexing.fusion.min_crossings = 2 on the fusion path. Caveat: the  
offline win is on the recall-oriented PROPOSAL candidate set; flipping the live served default  
stays owner-gated (production fusion segmenter is P2/unverified on real video).",  
    ]  
  }  
]
```

"GATE-A REAL-FOOTAGE CORROBORATION (same entry): offline re-score of GX010125_1080p's cached trajectory (no GPU): 45 candidates -> CONTROL 45 rallies vs Gate-A 27 rallies; Gate-A dropped 18 low-crossing windows (~1.7 min) of likely held-shuttle/feed FPs. No golden labels for this footage -> divergence, not F1.",

"GATE-A DATA SUPPORT (Data support for Gate A, and Standing lessons): real rallies on multi-court matches contain ≥ 2 net-crossings 97% of the time (median 5); a single service feed crosses ~once. So `min_crossings` ≥ 2 is data-backed.",

"LEARNED-VARIANT FIX #130 (docs/QUALITY_ITERATIONS.md, 2026-06-14, PR #130). Problem: the first A/B's learned logistic lost to CONTROL and zeroed testlarge ? a fixed-0.5-cutoff artifact. Fix: additive/default-OFF Variant.tune_threshold (operating point chosen on the TRAIN fold) + calibrate (Platt/isotonic, C2). Validated on n=6 golden set: mean F1 0.307 -> 0.423 (+0.116); testlarge 0.000 -> 0.300; the learned variant now beats CONTROL (0.388).",

"FIRST A/B LEARNED-LOGISTIC FAILURE (pre-fix, docs/QUALITY_ITERATIONS.md 'First end-to-end A/B' entry): the window-level 5-feature logistic did NOT clear the bar ? ?F1 -0.081, 2W/4L (a threshold/calibration artifact, not the stronger per-frame prototype). Same entry restates Gate-A: ?F1 +0.074, 6/6 matches, sign-test $p=0.031$, CI [+0.042, +0.104], over-segmentation 1.68 -> 1.01.",

"FUSION P3 (shuttle-only vs +person) ? DOC-DISPLAYED (docs/QUALITY_ITERATIONS.md fusion P3+P4 table, honest LOVO, IOU ≥ 0.5 , n=6). shuttle-only F1 0.307. +person (P3) F1 0.286, ?F1 -0.021 (-6.7%) vs prior, 95% CI [-0.165, +0.162], sign test 2W/4L ($p=0.688$), CI clears 0: No. Per-video: testlarge +0.40, gbaaddy -0.22 (a wash). Cause: no court polygons -> players_in_court counts spectators, so person adds noise.",

"FUSION P3 (shuttle-only vs +person) ? RAW JSON
(eval_baselines/experiment_leaderboard.json, first record, timestamp 2026-06-14T06:23:51Z, iou 0.5, label_set_hash 50d98ffbc370, num_videos 6). variant_a shuttle_only aggregate: precision 0.31807603780927995, recall 0.3101947096042718, f1 0.3068611057861412, mean_iou 0.5791831889148886. variant_b shuttle_person_fusion aggregate: precision 0.3617228001510626, recall 0.2708939086387243, f1 0.2863398846047612, mean_iou 0.5690111885714347. delta: precision +0.04364676234178266, recall -0.0393008009655475, f1 -0.02052122118138, mean_iou -0.01017200034345389. honest_delta_f1: mean_delta -0.02052122118137999, ci_low -0.16462806145542527, ci_high 0.16224446866060574, n 6, ci_excludes_zero false, sign_test wins 2 / losses 4 / ties 0 / n_effective 6 / p_value 0.6875.",

"FUSION P4 (shuttle+person vs +audio, incremental) ? DOC-DISPLAYED (same doc table). +person+audio (P4 incr.) F1 0.366, ?F1 +0.079 (+27.7%) vs prior (the shuttle+person 0.286 row), 95% CI [-0.054, +0.203], sign test 4W/2L ($p=0.688$), CI clears 0: No. Wins 4/6 (largetest +0.27, mahadevpura +0.27); shuttle+person+audio (0.366) lands above shuttle-only (0.307) but n=6 can't confirm.",

"FUSION P4 (shuttle+person vs +audio) ? RAW JSON
(eval_baselines/experiment_leaderboard.json, second record, timestamp 2026-06-14T06:24:56Z, iou 0.5, label_set_hash 50d98ffbc370, num_videos 6). variant_a shuttle_person aggregate: precision 0.3617228001510626, recall 0.2708939086387243, f1 0.2863398846047612, mean_iou 0.5690111885714347 (identical to P3's variant_b). variant_b shuttle_person_audio aggregate: precision 0.4530798177857001, recall 0.3408138120839734, f1 0.3655826552797436, mean_iou 0.7074802852140344. delta: precision +0.0913570176346375, recall +0.06991990344524907, f1 +0.07924277067498242, mean_iou +0.13846909664259965. honest_delta_f1: mean_delta 0.07924277067498244, ci_low -0.053508315003560535, ci_high 0.2029045239922802, n 6, ci_excludes_zero false, sign_test wins 4 / losses 2 / ties 0 / n_effective 6 / p_value 0.6875.",

"FUSION HONEST READING (doc): neither cue clears the promotion bar (both CIs span 0, both sign tests n.s.) at n=6 -> both stay default-OFF (promote-only-on-lift). Person adds noise (no court polygons -> players_in_court counts spectators); audio is the more promising lead (+0.079 / +27.7%, wins 4/6). Negative/uncertain results kept on purpose as the honest ablation backbone for the paper aim.",

"PRECISION DISCREPANCY TO FLAG FOR THE PAPER: the doc rounds the fusion F1 values (0.307, 0.286, 0.366) and ?F1 (-0.021, +0.079); the leaderboard JSON carries full float precision (shuttle-only 0.3068611057861412, +person 0.2863398846047612, +person+audio 0.3655826552797436; ?F1 -0.02052122118138 and +0.07924277067498242). The sign-test p in the doc table is written $p=0.688$ while the JSON stores p_value 0.6875 (same value, doc truncated). Use the JSON floats for any computation; do not re-round.",

"CORPUS SIZE. Golden corpus = the 6 labelled matches (n=6) in output/human_lovo_manifest.json (owner-confirmed 2026-06-13): adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles. All three eval JSONs and the leaderboard records report num_videos / n = 6. TOTAL RALLIES is NOT given as a corpus-wide aggregate in any file read; the only ground-truth rally counts present are per-video n_gt from the Gemini wrapper baseline: mahadevpura_singles n_gt=31, kushagra_singles n_gt=32 (these two videos sum to 63 GT rallies, but the other four videos' GT counts are not in

these files)."

```
],
"numbers": [
  "Heuristic F1 clean (Mahadevpura) = 0.657 | multi-court (Kushagra) = 0.157",
  "Gen-0 owned head LOVO n=6 F1 clean (Mahadevpura) = 0.744 | multi-court (Kushagra) = 0.561",
  "Two-stage WASB+Gemini refine F1 clean (Mahadevpura) = 0.83 | Kushagra = not measured (dash)",
  "Gemini wrapper (gemini-flash-latest) F1 clean (Mahadevpura) = 0.93 | multi-court (Kushagra) = 0.66",
  "Gemini wrapper Mahadevpura full: f1 0.93, precision 1.00, recall 0.87, mean_iou 0.80, boundary_err_s 0.7, n_gt 31, n_pred 27",
  "Gemini wrapper Kushagra full: f1 0.66, precision 0.69, recall 0.62, mean_iou 0.73, boundary_err_s 1.2, n_gt 32, n_pred 29 (contrast 1.4x)",
  "Heuristic LOVO floor mean_f1 = 0.480, std = 0.242 (n=6)",
  "Heuristic LOVO per-video F1: adarsh_avi_singles 0.751, kushagra_singles 0.157, targetest_doubles 0.737, gbaaddy 0.309, testlarge_short 0.267, mahadevpura_singles 0.657",
  "Gen-0 learned LOVO mean F1 (n=6) = 0.583 vs heuristic 0.480 (~+0.10); learned overtook heuristic at n=3; AUC 0.83-0.92",
  "Sign-test resolution rule (doc): need ~n=10 / 9 wins for p<0.05 (n=6 cannot resolve ~+0.08)",
  "Gate-A promotion: ?F1 = +0.074, 6/6 golden matches, sign-test p = 0.031, CI [+0.042, +0.104]",
  "Gate-A over-segmentation (segment-count ratio): 1.68 -> 1.01",
  "Gate-A recommended config: indexing.fusion.min_crossings = 2",
  "Gate-A real-footage (GX010125_1080p, no labels): 45 candidates -> CONTROL 45 rallies vs Gate-A 27 rallies; 18 windows (~1.7 min) dropped",
  "Gate-A data support: real rallies cross net >=2x 97% of the time, median 5; single feed ~1x",
  "First-A/B learned logistic (pre-fix #130): ?F1 = -0.081, 2W/4L (did NOT clear bar)",
  "Learned-variant fix #130: mean F1 0.307 -> 0.423 (+0.116); testlarge 0.000 -> 0.300; learned now beats CONTROL (0.388)",
  "Fusion P3 shuttle-only F1 (doc) = 0.307 | raw JSON = 0.3068611057861412 (precision 0.31807603780927995, recall 0.3101947096042718, mean_iou 0.5791831889148886)",
  "Fusion P3 +person F1 (doc) = 0.286 | raw JSON = 0.2863398846047612 (precision 0.3617228001510626, recall 0.2708939086387243, mean_iou 0.5690111885714347)",
  "Fusion P3 ?F1 (+person vs shuttle-only) = -0.021 (-6.7%) | raw JSON mean_delta -0.02052122118138; delta precision +0.04364676234178266, recall -0.0393008009655475, mean_iou -0.01017200034345389",
  "Fusion P3 honest stats: 95% CI [-0.165, +0.162] (raw [-0.16462806145542527, +0.16224446866060574]); sign test 2W/4L, p = 0.688 (JSON p_value 0.6875); ci_excludes_zero = false",
  "Fusion P3 per-video: testlarge +0.40, gbaaddy -0.22",
  "Fusion P4 +person+audio F1 (doc) = 0.366 | raw JSON = 0.3655826552797436 (precision 0.4530798177857001, recall 0.3408138120839734, mean_iou 0.7074802852140344)",
  "Fusion P4 baseline (shuttle+person) F1 = 0.286 (raw 0.2863398846047612), identical to P3's +person variant",
  "Fusion P4 ?F1 (+audio vs shuttle+person) = +0.079 (+27.7%) | raw JSON mean_delta 0.07924277067498242; delta precision +0.0913570176346375, recall +0.06991990344524907, mean_iou +0.13846909664259965",
  "Fusion P4 honest stats: 95% CI [-0.054, +0.203] (raw [-0.053508315003560535, +0.2029045239922802]); sign test 4W/2L, p = 0.688 (JSON p_value 0.6875); ci_excludes_zero = false",
  "Fusion P4 per-video wins (4/6): targetest +0.27, mahadevpura +0.27",
  "Corpus size: n = 6 golden videos (num_videos=6 in all records). Total rallies: NOT aggregated anywhere; only per-video n_gt present are Mahadevpura 31 and Kushagra 32 (sum 63 for those two videos only; other 4 videos' GT counts absent)",
  "Both fusion leaderboard records: iou 0.5, label_set_hash 50d98ffbc370, num_videos 6; P3 timestamp 2026-06-14T06:23:51Z, P4 timestamp 2026-06-14T06:24:56Z"
],
"source_refs": [
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\QUALITY_ITERATIONS.md (lines 25-37 baselines table; lines 43-60 fusion P3+P4 entry; lines 62-83 Gate-A PROMOTED + #130 fix; lines 85-94 first-A/B entry; lines 180-188 archived multivideo-lovo predecessor; lines 192-212 standing lessons incl. net-crossing 97%/median 5 and sign-test n-rule)",

```

```

"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\eval_baselines\\heuristic_lovo_n6.json (mean_f1 0.480, std 0.242, per_video F1 map;
computed 2026-06-10 via backend.eval.calibrate_local)",
"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\eval_baselines\\gemini_wrapper_2026-06-10.json (per_video mahadevpura/kushagra
f1/precision/recall/mean_iou/boundary/n_gt/n_pred; comparison_at_time_of_run heuristic_lovo +
gen0_learned_lovo; model gemini-flash-latest)",
"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\eval_baselines\\experiment_leaderboard.json (record 1: shuttle_only vs
shuttle_person_fusion, P3; record 2: shuttle_person vs shuttle_person_audio, P4 ? full float
aggregates, deltas, honest_delta_f1 CIs and sign_test)",
"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\output\\human_lovo_manifest.json
(named as the n=6 golden corpus source in QUALITY_ITERATIONS.md line 18; not opened in this pass
? the 6 match names are quoted from the doc)"
]
},
{
"area": "SYSTEM + METHOD + measurement-honesty framework (rally detection)",
"facts": [
"2-LAYER MENTAL MODEL (docs/HOW_RALLY_DETECTION_WORKS.md): Layer 1 = DETECTION ('where is
the shuttle this frame?') = WASB CNN, a FROZEN shuttle-trajectory substrate emitting per-frame
(x,y,visible). Shuttle is chosen as the camera-INVARIANT primary because 'shuttle in flight' =
'a rally is happening', unlike player-motion which fooled the old heuristic on broadcast pan
cameras. Layer 2 = SEGMENTATION ('when is a rally?') = compute shuttle VELOCITY from the per-
frame (x,y), then turn it into [start,end] rally windows.",
"THE SUBSTRATE->CANDIDATE-WINDOWS->SCORER CHAIN: frames -> WASB frozen trajectory ->
velocity signal -> shuttle-seeded CANDIDATE WINDOWS
(TrackNetRunner.trajectory_to_action_windows, the model-agnostic 'windowing brain') -> a LEARNED
SCORER weights per-window feature columns -> [start,end] rallies graded by temporal IoU vs
golden labels. Original v0 segmentation = 3 hand-tuned knobs (velocity_thresh, merge_gap,
min_window_duration); the learned scorer replaces hand-tuning.",
"GOLDEN LABELS GRADE/TUNE LAYER 2 ONLY: Roora/golden labels are rally [start,end] TIMES
(when), not (x,y) coordinates (where), so they cannot retrain WASB's frozen detector; they grade
and tune the segmentation layer. Honesty rule baked in: 'calibrate on some rallies, measure on
held-out ones' (LOVO).",
"SIGNALS ARE CONFIDENCE-WEIGHTED FEATURE COLUMNS, NOT BRANCHES
(docs/ANALYZERS_AND_RECONCILERS.md §1, the spine): every producer (detector/analyzer/reconciler)
emits a SIGNAL = (value, confidence/presence) -> persisted to candidate_segments.stats as a
namespaced column '<producer>__<key>' (+ '_confidence'). The SCORING MODEL consumes those
columns and LEARNS TO WEIGHT them. NO hand-coded if/else branches in the decision path.
'Producers propose; the scorer arbitrates; the experiment harness + golden set decide what gets
promoted.'",
"THE SCORER: a tiny pure-numpy L2 LogisticRegression (backend/eval/classifier.py, ADR-004)
-- deliberately NOT a neural net, to survive small-N. _gate_mask (threshold a single column) is
the trivial scorer; LogisticRegression (weight all columns) is the learned scorer. A signal that
doesn't earn its keep stays at weight ~0 -- it costs nothing because it is an unused column, not
a live if-branch.",
"NO SPECIAL-CASING / DEFAULT-OFF / PROMOTE-ONLY-ON-LIFT discipline
(docs/EXPERIMENT_HARNESS.md): experimentation and deployment are DECOUPLED. (1) Merge
experimental code freely -- it is gated OFF (per-cue enable flag in IndexingConfig, default
False) so it cannot change production behavior. (2) Test offline against golden labels (most
experiments are pure re-scoring of persisted stats -- never touch the served path). (3) Promote
a variant to default ONLY on measured held-out LOVO lift, through a CI eval gate
(run_regression.py, exit 1 blocks). (4) 'Rollback = flip a config flag, never git revert' -- a
pinned/frozen CONTROL variant (recipe + classifier model hash) is always registered and is the
default.",
"DETECTION-perp-DECISION SUBSTRATE (the persist-once/re-score-forever property): expensive
GPU/WSL DETECTION (shuttle WASB trajectory, person tracks/boxes, audio hits) runs ONCE per video
and is persisted (candidate_segments.stats JSON); cheap pure-Python DECISION (windowing + gates
+ the LR scorer) re-scores any number of variants OFFLINE, deterministically, with no GPU and
zero production risk. This is what makes every experiment a zero-risk offline re-score.",
"FUSION = TWO MODES (docs/MULTI_SIGNAL_FUSION_PLAN.md §3): (3.1) DECISION-LEVEL GATE
(label-free, ships first): rally_active = shuttle_motion AND (>=2 persons in court OR recent
person track); composes Gate A (net_crossings>=2, the held-shuttle/service-feed structural fix,
EXISTS in rally_gate.py but was eval-only -- wired to prod in fusion P2) AND Gate B (the person

```

cue, NEW). (3.2) FEATURE-LEVEL (the learned path, primary): cues emit a SMALL set of aggregated, SCALE/TRANSLATION-INVARIANT per-window features (counts/ratios/zone-membership/two-sidedness -- NEVER raw coordinates, which would memorize this court/camera) that join the LR feature vector.",

"SHUTTLE STAYS PRIMARY, person is gate+feature never the sole trigger; single-cue segmenters remain registered fallbacks (fusion-degradation guard). Person runs ONLY over shuttle-seeded candidate windows + padding, not the whole video (bounds cost). ~4-5 person features deliberately (more features on a small labeled set overfit -- the same reason the model stays an LR, not a net).",

"THREE TEMPORAL SCOPES of producers (docs/ANALYZERS_AND_RECONCILERS.md §2): frame-scope detectors (RallyCue, expensive GPU/WSL: WASB shuttle, person boxes, audio hits, shuttle-state), window-scope WindowAnalyzer (cheap/pure: service-fault, double-bounce, let-cord), session-scope SessionAnalyzer (cheap/pure: warm-up/practice/break spans). Coarser-scope signals aggregate finer ones; all ultimately land as per-candidate feature columns. The scorer is the ONLY place a decision is made and it does so by weighting columns, never branching.",

"RECONCILERS = the BACKWARD seam, a 'linter for rally decisions' (§6): given predicted segments + persisted evidence, detect where the DECISION contradicts the EVIDENCE (trailing_dead_time->truncate, no_net_crossing->drop, over_merged->split, boundary_slack->tighten, inside_warmup->flag) and propose corrections. REPORT-FIRST + apply-optional (never silently rewrites; default surfaces a report for human review; flagged reconcile.apply records every change). Idempotent (reconcile(reconcile(x))==reconcile(x)), fixed rule precedence. It is a Variant transform so its lift is A/B-measurable BEFORE auto-apply.",

"MEASUREMENT-HONESTY CORE: LEAVE-ONE-VIDEO-OUT (LOVO), group by MATCH -- train on the OTHER videos' candidates, predict the held-out VIDEO, score vs its labels; NEVER a held-out window from the same video (windows in one video are correlated -> flatters the score). The field's name = leave-one-group-out / leave-one-subject-out (LOGO/LOSO) grouped CV. distill_local.py already implements this; calibration.leave_one_video_out is the primitive.",

"TRAINING LABELS vs MEASURING STICK: today's training labels are GEMINI SILVER labels (distilled offline, ADR-004/007) -- Gemini is a TEACHER, never a runtime dependency and NEVER a training-data SOURCE for owned weights (guardrail). GOLDEN labels (Roora vendor) are the HONEST measuring stick; as the golden corpus grows they can also become training labels. Until the labeled set is larger, learned weights are treated as 'calibration / direction-check, not a fully general model'.",

"HONEST RELATED-WORK FRAMING (docs/ANALYZERS_AND_RECONCILERS.md §13, literature review 2026-06-14, multi-agent web research, every citation adversarially verified -> ZERO fabricated papers across ~40 checks). TL;DR: the architecture's SKELETON is STANDARD and must be CITED, NOT CLAIMED. Candidate-windows + learned scorer = Temporal Action Localization (TAL, proposal-then-score). Reconciler's over-segmentation fixes = Temporal Action Segmentation (TAS) refinement/post-processing. 'Signal not verdict / learned arbiter over cue columns' = late/score-level fusion realized as stacked generalization (~20 years old, ALREADY present in racket sports).",

"WHAT IS ACTUALLY NOVEL (§13.1/§13.3): of the patterns surveyed, ONLY the report-first RECONCILER-AS-LINTER bundle rated 'novel-combination' (no found paper bundles report-first contradiction-detection over a persisted temporal decision + auditable structured interval corrections + per-rule A/B lift-gating BEFORE auto-apply + idempotency/precedence + human-in-the-loop + feeding a (value,confidence) scorer). All other patterns = 'partial-overlap'. The thinnest other genuine gap = warm-up-as-a-soft-session-feature (novel only by absence of counterexample). 'Cite, don't claim' is the operating rule; park any novelty claim narrowly on those two.",

"WHAT CANNOT BE CLAIMED (§13.3): the fusion mechanism (multi-cue features -> light scorer) is ~20 yrs old in racket sports (CVIU 2009; Ghosh et al. 2018 SVM-over-features); the persist-once/re-score substrate is the DEFAULT of the TAD field (claim only the idempotent/versioned discipline); the cross-check-and-correct insight already exists (Hsu et al. 2024 cross-check shuttle-vs-swing; EventAnchor does human-in-the-loop suggest-then-verify -- our delta is measured-before-apply + auditability); static single-camera as a novel setting and the no-regression/flag-gated harness as a primitive (textbook MLOps) -- claim only 'this discipline on a deliberately tiny golden set'.",

"IN-DOMAIN PRIOR ART hard-codes our cues as if/else verdicts (so our delta is the NO-BRANCH design philosophy, not the cues): non-broadcast badminton rally start/end (See et al. 2024/25), shuttle-state/hit (MonoTrack arXiv:2204.01899; Hsu/Yu/Cheng Sensors 2024), service-fault (Goh et al. Sensors 2023), tennis in-play state machines, badminton point-segmentation (Ghosh et al. 2018 arXiv:1712.08714), ShuttleSet (arXiv:2306.04948)."

],

"numbers": [

"C1 -- BIAS-AWARE LOVO + per-fold CIs (Tier-1, do first): standard leave-one-out on ~6

imbalanced groups induces a train-vs-heldout label anti-correlation that biases the estimate (a cited LR scored ~0.23 auROC on RANDOM data). Fix: class-balance-preserving folds (RL00CV) and REPORT PER-FOLD VARIANCE + CIs, never a single mean, so the promotion gate can't promote noise. Source: Austin/Pe'er/Korem 2025, arXiv:2406.01652 (Science Advances 10.1126/sciadv.adx6976). SHIPPED: backend/eval/eval_stats.py provides bootstrap 95% CI + paired sign test on the fusion-lift; compare(..., honest_stats=True) attaches a 'honest' block (per-metric bootstrap CIs + paired DeltaF1 CI + exact sign test) -- commits #124 (C1-C4 helpers) and #127 (honest DeltaF1: bootstrap CI + paired sign test on the fusion lift).",

"PROMOTION RULE (encoded in EXPERIMENT_HARNESS §5.1): a bare LOVO mean at n~6 is NOT promotable on its own; the lift is real ONLY WHEN the DeltaF1 CI clears 0 AND the sign test agrees. honest_stats defaults ON and is ADDITIVE (existing output unchanged; honest_stats=False restores legacy shape).",

"C2 -- CALIBRATE EACH PRODUCER'S CONFIDENCE (Platt/isotonic) BEFORE the LR, so a CNN softmax, an audio onset, and an integer net-crossing count are comparable and the over-confident cue can't swamp the arbiter. Source: MoCaE arXiv:2309.14976; modality competition arXiv:2203.12221.",

"C3 -- OUT-OF-FOLD (honest) STACKING: when a producer is itself learned (shuttle-state), feed the LR its OUT-OF-FOLD outputs; extend the group-by-match split to producer training (avoids in-sample leakage). Source: Wolpert 1992.",

"C4 -- REPORT THE FIELD'S METRICS: add SEGMENTAL F1@{10,25,50} + EDIT SCORE (TAS) and a TIGHT/LOOSE mAP@delta sweep + SEGMENT-COUNT RATIO alongside temporal-IoU, because tIoU is nearly BLIND to over-segmentation -- a reconciler win (or an IoU-chasing rule that shreds rally structure) is invisible under tIoU alone. Source: MS-TCN; ASRF; SoccerNet arXiv:1804.04527; Soares 2022 arXiv:2206.07846. SHIPPED helper: backend/eval/segmentation_metrics.py.",

"Tier-2 corrections: C5 resolve reconciler rules GLOBALLY (activity-grammar/constraint-aware Viterbi) not stacked local edits; C6 brand reconciler as 'model/consistency assertions' + per-correction confidence + margin-ranked human queue (HoloClean ~90% repair precision => ~10% bad); C7 a few EARLY-fusion interaction columns (e.g. shuttle_in_air AND person_two_sided) since a linear arbiter can't represent AND/XOR; C8 represent a missing cue as explicit is_present/uncertainty column NOT a silent 0; C9 keep warm-up a SOFT feature never a hard pre-filter + court/context-invariance probe (action-context confusion arXiv:2103.16155).",

"Tier-3: C10 model cue CORRELATIONS (Snorkel/MeTaL + strong L1/L2 -- net-crossing/two-sidedness/shuttle-in-air co-fire so an unregularized LR double-counts); C11 a temporal-smoothing pass (Kalman/HMM/Viterbi or MS-TCN truncated-MSE anti-fragmentation penalty) before cutting intervals (SmartTennisTV F1~97.5% largely via Kalman-smoothing); C12 pull ShuttleSet/ShuttleSet22 + CoachAI as external benchmark/weak-supervision corpus to attack the ~6-match bottleneck; C13 benchmark the rule reconciler against a learned boundary refiner (ASRF/CBR) to justify staying rule-based.",

"FOUR NO-REGRESSION INVARIANTS (the load-bearing design claims, §2): (1) Signal not verdict; (2) Expensive detection PERP cheap decision; (3) Report-first reconciliation; (4) Promote only on measured held-out lift (LOVO + run_regression.py, default OFF, rollback=config flip). KEYSTONE TEST: all analyzer flags OFF + reconciler report-only => candidate rows AND predictions BYTE-IDENTICAL to control.",

"CORPUS SIZE = 6 matches (QUALITY_ITERATIONS) -- enough for direction-checks and LOVO sign-tests, NOT enough to claim a general model; corpus growth is the highest-ROI unblock. Current learned-classifier feature set (distill_local.py FEATURE_NAMES): duration, peak_velocity, mean_velocity, active_ratio, net_crossings (net_crossings already a feature). Person features to add: ~4-5 (players_in_court, two_sided_motion, mean_player_motion, in_court_ratio, shuttle_player_colocation -- the direct held-shuttle discriminator).",

"RISK REGISTER (§13.5, pitfalls actively being walked into): over-segmentation hidden by tIoU (->C4); late-fusion miscalibration/dominant-cue swamping (->C2); LOOCV distributional bias + tiny-N variance (->C1); stacking leakage + threshold-tuning-on-test (keep EVERY tunable knob inside the LOVO loop); proposal-recall ceiling (decision can't recover a window the windowing missed -- measure AR@AN separately); report-first decaying into silent rewrite; static-tripod transfer gap (broadcast cues don't exist on our footage); detector domain gap inherited downstream (WASB/TrackNet trained on other rigs -- QUANTIFY detector accuracy on our footage before trusting any cue column; currently UNMEASURED)."

],

"source_refs": [

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\HOW_RALLY_DETECTION_WORKS.md (2-layer mental model: DETECTION frozen WASB shuttle substrate -> velocity -> SEGMENTATION 3 knobs -> [start,end] windows graded by temporal IoU; golden labels grade/tune layer 2 only; 'calibrate on some, measure on held-out')",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\MULTI-SIGNAL-FUSION_PLAN.md (§2 one rally-fusion layer, cues = swappable RallyCue

producers, shuttle primary/camera-invariant; §3.1 decision-level Gate A AND Gate B; §3.2 feature-level learned path, scale/translation-invariant feature columns, the numpy LR scorer ADR-004; §3.2 small-N discipline LOVO + Gemini silver vs golden; §4 in-flight-vs-held discriminators; P2 DONE 2026-06-14 fusion_hybrid default-OFF)",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\EXPERIMENT_HARNESS.md (§2 detection perp decision / persist-once re-score-forever; §3 audit of existing eval stack incl. calibration.leave_one_video_out, regression.regress, classifier.LogisticRegression; §4 Variant = declarative recipe, pinned CONTROL; §5 three modes incl. §5.1 compare honest_stats block; §6 four no-regression guarantees, default OFF, rollback=config flip)",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\ANALYZERS_AND_RECONCILERS.md (§1 the spine: signal not verdict, no special-casing, scorer weights columns; §2 three scopes + four invariants; §3 forward seam analyzers incl. service-fault + warm-up examples; §6 backward seam report-first idempotent reconciler-as-linter; §10 generality + honesty caveat; §13 FULL literature review -- §13.1 verdict table cite-don't-claim + only reconciler novel-combination, §13.2 terminology gloss, §13.3 can/cannot claim, §13.4 course-corrections C1-C13, §13.5 risk register, §13.6 residual gaps)",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\CODE_MAP.md (§1A-1C runtime + WASB substrate + calibration/eval flow; §2.1 segmenters incl. trajectory_hybrid shared engine + fusion_hybrid Gate A/B; §2.2 detectors incl. tracknet_runner trajectory_to_action_windows 'windowing brain', person_detector, fusion_features held-shuttle discriminator, rally_gate Gate A)",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\eval\\eval_stats.py and backend\\eval\\segmentation_metrics.py (SHIPPED C1/C4 measurement-honesty helpers -- bootstrap CI + paired sign test, segmental F1@k + edit + segment-count ratio; commits #124 measurement-honesty helpers C1-C4 additive/default-OFF, #127 honest DeltaF1 bootstrap CI + paired sign test on the fusion lift)")

]

},

{

"area": "PLANNED work + corpus/labeling story (owned-model roadmap, execution-policy resilience, next experiments, Roora-as-oracle)",

"facts": [

"OWNED-MODEL ROADMAP (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) ? status is PLAN/roadmap only; no owned-model implementation starts until the owner greenlights a specific milestone, and paid Gemini stays the always-ready inference/fallback throughout, never a training source.",

"North Star: immediate deliverable = a training framework + harness producing a high-precision/recall highlight+rally extractor (quality is the first-order objective); eventual product = a single sport-agnostic, conversational video model (upload clip, ask 'longest rally', 'all service faults', 'rallies > 20 shots'); on-device is LATER (compression goal). The moat = consented dataset + eval harness, NOT the weights.",

"BASE MODEL: Qwen3-VL (Apache-2.0, uniform across sizes) is primary ? long-video context, purpose-built second-level timestamp emission, multi-turn video chat. InternVL3 (MIT/Apache) is the clean fallback if a Qwen blocker appears. Gemma-4 stays AMBER / bench-only (Apache grant likely but Prohibited-Use posture unconfirmed, counsel needed; video/timestamp capability disputed). Reject Gemma 3 (viral), Gemma 3n, Llama 4 (700M-MAU cap), Commons-Clause repos.",

"FRAMEWORK (not the base): ms-swift (Apache-2.0) is the chosen forge ? native video + audio + timestamp-grounding + GRPO/RFT with a custom video reward in one stack. unsloth (Apache-2.0 core; avoid its AGPL Studio UI) is kept as the 8GB-local SFT/compression tool for the deferred on-device goal.",

"DECISIVE SEQUENCING CALL: start with the TAL (temporal action localization) head, NOT the VLM, because fine-tuned VLMs miss strict temporal boundaries and boundary accuracy IS the product. The head (rally_seq_proto.py) is already de-risked (held-out AUC 0.83-0.92), trains in minutes at ~4.5GB on a 2070 Super (\$0), is ~1M params, and doubles as a pseudo-label teacher + the honest baseline the VLM must beat. Strict order: head now -> VLM later; when the VLM comes, prefer two-stage (VLM coarse-proposes, head refines boundaries) over wholesale replacement.",

"HARD LICENSING GUARDRAIL (non-negotiable): commercial-clean licensing for EVERY dependency ? code, data, AND model weights ? MIT / Apache-2.0 / BSD only. Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs; also forbidden are public grounding-CoT datasets that were Gemini-generated (e.g. TVG-R1's 56K) ? the project must originate its own CoT supervision. Exclude minors; per-user training data needs separate explicit opt-in; split data BY MATCH. (Caveat: base-model pretraining provenance is unauditably for every open VLM incl. Qwen3-VL ? needs an explicit owner risk-acceptance.)",

"CONVERSATIONAL VIDEO-QA: build the bookkeeping NOW, punt the model. Architecture = 'extract once, query many': EXTRACTION (TAL head -> later VLM) emits structured timestamped spans -> a durable per-video EVENT STORE -> a structured QUERY layer (text-to-SQL + ffmpeg clip-

cut). Cited queries are structured-numeric -> SQL, not a VLM job (SQL counts exactly where VLMs hallucinate). Reserve the event/fault-tag taxonomy (e.g. service_fault) up front; punt the NL parser/agent until extraction quality clears the bar.",

"PHASE 0 (start now, per-milestone greenlight, \$0/local): M0.1 = the labeled-corpus spine + event-index contract (the moat, canonical JSONL row per rally wired to a store; authored in sports-data-collector as system-of-record); M0.2 = grow the corpus (DONE n=6, running); M0.3 = compliance cleanup (HARD BLOCKER ? purge the Gemini-silver training path in backend/eval/distill_local.py + calibrate_local.py; resolve WASB C3 checkpoint-provenance via own-trained weights or a clean MIT/Apache detector swap); M0.4 = Gen-0 TAL harness (productionize rally_seq_proto.py -> one-command train->eval->ship-gate over ActionFormer (MIT) / ASFormer (MIT)). Phase 1 = Gen-0 ship decision (gated on n>=5 + M0.4). Phase 2 = Gen-1 (concatenate local cue channels) / Gen-2 (ms-swift Qwen3-VL fine-tune, SFT cold-start -> GRPO/RFT with temporal-IoU-vs-golden reward, own CoT supervision, two-stage), gated on platform scaffolding + healthy corpus + explicit go.",

"EXECUTION-POLICY RESILIENCE FRAMEWORK (docs/EXECUTION_POLICY.md, DESIGN ? owner-gated, NOT started): motivated by a real GX010125 hang where a Gemini generateContent call stalled with NO client-side timeout and blocked the whole run. The fix is a per-job declarative ExecutionPolicy (just JSON data riding on the job row), not a one-off timeout ? every long operation declares how it waits/polls/aborts/resumes.",

"ExecutionPolicy contract: WaitMode enum = WAIT_AND_RESUME (DEFAULT ? bounded wait; on deadline persist+reschedule), ABORT (stop the op, fail or partial), BLOCK (legacy in-process blocking wait). Fields: poll_every_n_sec=10.0, op_timeout_sec=120.0 (the HARD per-attempt deadline ? the hang-killer that was missing), max_wait_sec=1800.0 (total wall budget across resumes), max_retries=5 (failures, exponential backoff+jitter), backoff_base_sec=3.0, on_timeout in {reschedule,partial,fail}, resumable=True. Hang vs failure are distinct: max_retries+backoff handle a call that ERRORS, op_timeout handles a call that HANGS.",

"SELF-SCHEDULING (no master node): a job under WAIT_AND_RESUME runs the external call under op_timeout_sec; on timeout/in-progress it persists progress + sets next_poll_at = now + poll_every_n_sec, sets job status=waiting, and RELEASES the worker (no thread sleeping for minutes) ? re-enqueuing itself. The existing single ThreadPoolExecutor worker (main.py, max_workers=1) re-claims due jobs; the crash-recovery sweep (fail_stale_jobs, DEFERRED_HARDENING #16) reaps dead-process jobs. The DB IS the coordinator ? needs only two jobs-table columns (next_poll_at, progress JSON already exists) + a claim-due query, so it generalizes to N workers with no scheduler/leader/master. Phasing P0 (doc) -> P1 (ExecutionPolicy + bound the two Gemini waits) -> P2 (per-job policy + self-scheduling) -> P3 (resumable AI handoff, per-window confirmation) -> P4 (same policy on WSL/WASB GPU + ffmpeg).",

"CONCRETE NEXT EXPERIMENTS (docs/NEXT_STEPS.md) ? #1 highest-leverage: GROW THE GOLDEN CORPUS. Priority order: (a) multi-court badminton (target distribution AND where the ship target lives), (b) >=3 matches per new sport, table tennis first (WASB transfers at F1 0.909; pickleball labels are corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety; adult sessions ONLY for the training pool. Per video: label -> curate import-local --normalize --license Proprietary --fps <true fps> -> re-run the Gen-0 harness. The paired sign test needs n: at n=10 with 9 wins p~0.011 ? significance is reachable this month.",

"NEXT EXPERIMENT ? ACTIVATE COURT POLYGONS so the person/court-bounds cue stops counting spectators: the FusionSegmenter (fusion_hybrid.py, default-OFF, P2 DONE 2026-06-14) already supports an optional court mask (fusion.court_polygon) and Gate A net-crossings + optional Gate B min_players; the Roora intake manifest now collects court_corners + net_line + singles_or_doubles per video so the detector can ignore people OUTSIDE the court, require >=2 players ON court for a rally, and count shuttle net-crossings. Next after P2 is P3 (learned fusion: person features -> harness compare for a LOVO lift on the 6 golden videos).",

"NEXT EXPERIMENT ? MEASURE AUDIO ON SHUTTLE-ONLY DIRECTLY: audio is parked (E1 probe PR #35 found it weak ? ~2.2x discrimination, AUC ~0.6 ? and the foreground-audio idea also failed); it is NOT 'in reserve.' Pose/court is the more credible #2 cue. The planned audio path is Fusion P4: an audio cue via backend/pipeline/audio/hit_detector.py, landing flag-gated/default-OFF and promoted only on a measured LOVO lift.",

"NEXT EXPERIMENT ? FUTURE 'SHUTTLE-DROPPED' CUE: the cheapest current quality win is wiring the already-built net-crossing Gate A (backend/pipeline/detectors/rally_gate.py, apply_gate(..., min_crossings=2)) into the live segmenter ? it kills service-feed/held-shuttle windows but is currently called only from eval drivers (distill_local.py, eval_partial_labels.py, export_wasb_segments.py, calibrate_local.py) with no min_crossings knob in IndexingConfig, so production never applies it. The roadmap's faithful serve-CONTACT boundary definition (start = contact, not the toss/hold) is what structurally fixes the 'held-shuttle counted as a rally' false positive; a future 'shuttle-dropped = shuttle-on-floor for a contiguous frame sequence' style cue extends this same shuttle-substrate reasoning.",

"EXPERIMENT DISCIPLINE: every new cue lands flag-gated, default OFF; promote to default ONLY on a measured LOVO lift through the run_regression.py gate (exit 0/1/3); a pinned CONTROL variant means rollback = config flip, never git revert. The experiment harness (backend/eval/experiment.py, P1 DONE) is ~80% existing code, wrapping calibration.py/metrics.py/classifier.py/training.py/regression.gt_hash + the DB bridge.",

"ROORA AS AN ORACLE LABEL GENERATOR: the labeling vendor (Roora, HSR Layout/Bangalore, info@rooragr.com) is treated as an interchangeable oracle that emits trustable ground-truth golden labels. The system's contract is with the golden LABELS (assumed-trustable ground truth, validated by automated checks + ~20% human spot-check at ± 0.3 s boundary tolerance), NOT with their provenance ? so evaluation rigor (metrics.match_intervals, LOVO, the ship-gate) is decoupled from the vendor, and the corpus scales by ADDING or SWAPPING oracles (Roora's labels import directly via 'python -m src.cli.curate import-local' in the same schema any oracle must satisfy). The canonical guideline lives in the sibling sports-data-collector repo (docs/annotation/); this repo's copy is a mirror.",

"CORPUS = OWNER-GROWN, PROPRIETARY, MINORS EXCLUDED: the n=6 owned corpus (262 rallies) lives in the collector DB as system-of-record, classified Proprietary (commercial export = owned data only after the ShuttleSet reclassification), grown by the owner. Hard data guardrails enforced at the labeling/contract layer: exclude minors entirely (footage with children does NOT enter the labeled pool); no biometric/identity labeling (per-video pseudonyms only ? 'near side'/'far side', never real names or cross-video person IDs); no gait/pose/skeleton/keypoint annotation; work-for-hire IP assignment so labels are owned work product; data split BY MATCH; per-user training data needs separate explicit opt-in."

```
],
"numbers": [
  "Qwen3-VL = Apache-2.0 (primary base); InternVL3 = MIT/Apache (fallback); ms-swift
framework = Apache-2.0; unsloth core = Apache-2.0",
  "Licensing whitelist: MIT / Apache-2.0 / BSD only (code + data + weights)",
  "Forbidden Gemini-generated CoT example: TVG-R1's 56K dataset",
  "TAL head (rally_seq_proto.py): held-out AUC 0.83-0.92; trains in minutes at ~4.5GB on a
2070 Super; ~1M params; $0",
  "Best RL-post-trained video VLM ~ R@0.7 50% on open-domain benchmarks (why head-first)",
  "ExecutionPolicy defaults: poll_every_n_sec=10.0, op_timeout_sec=120.0,
max_wait_sec=1800.0, max_retries=5, backoff_base_sec=3.0",
  "Gemini hang shape: file-processing poll = while state==PROCESSING: sleep(10) (no
deadline); generate_content retry loop = 6 attempts (catches failures not hangs)",
  "Worker: ThreadPoolExecutor max_workers=1; self-scheduling needs only 2 jobs-table columns
(next_poll_at + existing progress JSON)",
  "Gen-0 owned head LOVO (n=6) = 0.626 (artifact v0.1.0; floor passed; sign test 5/6 wins
p=0.109 -> not significant, ship=False)",
  "Quality table @IoU 0.5 vs human golden ? Heuristic: Mahadevpura 0.657 / Kushagra 0.157;
Gen-0 head (LOVO): 0.744 / 0.561; two-stage WASB+Gemini: 0.83 / ?; Gemini wrapper: 0.93 / 0.66",
  "Ship-gate references: floor = heuristic LOVO 0.480 (necessary not sufficient); multi-
court reference = Gemini wrapper 0.66 (the owned-model ship target); clean-footage ceiling =
0.93. Suggested target: LOVO F1@tIoU0.5 >= 0.70 on multi-court folds, paired-sign p<0.05 vs
heuristic",
  "Significance path: n=10 with 9 wins -> p=0.011 (reachable this month); current n=6",
  "Gate A: rally_gate.apply_gate(..., min_crossings=2)",
  "Audio E1 probe (PR #35): ~2.2x discrimination, AUC ~0.6 (parked, not adopted)",
  "WASB transfer to table tennis: F1 0.909",
  "Owned corpus: n=6 matches / 262 rallies, Proprietary; Roora boundary tolerance  $\pm 0.3$  s
(~ $\pm 9$  frames @30fps); shots_count exact for rallies  $\leq 15$ ,  $\pm 1$  for longer",
  "Roora phasing: Phase A pilot = 3 videos (1 badminton + 1 pickleball + 1 table tennis), <3
min each, ~10-12 rallies; Phase B scale = ~12-15 videos/sport (~36-45 total), ~30 min each",
  "Court calibration manifest: court_corners = 4 points clockwise from near-left; net_line =
2 endpoints (left post -> right post); singles_or_doubles sets expected on-court player count (2
vs 4)"
],
"source_refs": [
  "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\OWNED_MODEL_IMPLEMENTATION_PLAN.md",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\NEXT_STEPS.md",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\EXECUTION_POLICY.md",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\ROORA_LABELING_GUIDELINES.md",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-
```

```

indexer\\docs\\MULTI_SIGNAL_FUSION_PLAN.md (referenced)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\EXPERIMENT_HARNESS.md
(referenced)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\backend\\pipeline\\detectors\\rally_gate.py (Gate A, eval-only)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\backend\\pipeline\\detectors\\fusion_hybrid.py (FusionSegmenter, default-OFF)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\eval\\experiment.py
(experiment harness)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\eval\\distill_local.py
+ calibrate_local.py (Gemini-silver training path to purge, M0.3)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\training\\gen0\\harness.py
(Gen-0 one-command harness)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\backend\\providers\\gemini.py::analyze_video (the GX010125 hang site)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\features\\rally_seq.py
(single-sourced featurizer, FEATURE_SCHEMA v1)"
]
},
{
  "area": "Engineering + reproducibility: replayable feature substrate, experiment
leaderboard, run telemetry black box, default-OFF + CI-gated discipline",
  "facts": [
    "PERSIST-ONCE / RE-SCORE-FOREVER SUBSTRATE: fusion_golden.py runs the GPU step exactly
ONCE per video. It reuses EXISTING golden trajectory CSVs (no WASB re-run), needs no Gemini and
no DB, and the GPU touches exactly one thing -- PersonDetector person detection over the
original MP4. extract_video() emits one small JSON per video ({name, fps, frame_width,
candidates:[{start,end,shuttle{,person{,label}], gts}) where shuttle = 5 fps/resolution-
invariant shuttle features + person = ~5 person-cue features per candidate window. The module
docstring states each output_<name>_golden_features.json is 'the replayable feature substrate --
re-run fusion_compare with different variants/thresholds forever without re-detecting on the
GPU.'",
    "DETECTION vs DECISION SPLIT is the reproducibility thesis (EXPERIMENT_HARNESS.md §2,
restated in experiment.py header): separate the expensive, risky, non-deterministic per-frame
DETECTION (run once on GPU/WSL, persisted into candidate_segments.stats / the golden JSONs) from
the cheap, swappable, DETERMINISTIC DECISION (given those signals, where are the rallies). A
Variant is a declarative re-scoring recipe (NOT a code branch); given persisted features it
produces List[Interval] deterministically with no GPU and zero production risk. 'No new scoring
math lives here' -- experiment.py composes existing modules (metrics, training, classifier,
gemini_refine, regression, calibration).",
    "EXPERIMENT LEADERBOARD: record_experiment() appends one compact reproducible row per
compare() to eval_baselines/experiment_leaderboard.json -- a TRACKED location (NOT gitignored
output/), so the leaderboard survives a clean checkout (mirrors
regression.DEFAULT_BASELINE_PATH). Each row carries variant_a/variant_b names + spec_hash +
aggregate metrics + delta, and (when present) honest_delta_f1 (CI + sign test). timestamp is
injectable for deterministic tests.",
    "REPRODUCIBILITY VIA HASHING: Variant.spec_hash() is a stable 12-char sha1 of the recipe
dict -- 'makes a result reproducible. The pinned control variant always hashes the same, so a
regression is always traceable to a recipe change, never to drift.' compare() also records
label_set_hash = gt_hash(all golden intervals), which 'detects the deltas were measured against
different labels' the same way regression.gt_hash guards the no-regression baseline. So every
leaderboard row is pinned to (exact recipe, exact label set, exact IoU).",
    "HONEST MEASUREMENT (not bare means): learned variants are scored LOVO (Leave-One-Video-
Out) -- train on the OTHER videos, predict the held-out one (the number to trust); a pinned
classifier_model scoring its own training videos is explicitly flagged optimistic.
honest_summary adds bootstrap CIs per metric, a paired ?F1 CI + exact sign test (C1), and F1@k +
segment-count ratio (C4, the over-segmentation tIoU hides). The threshold/calibrator are fit on
the TRAINING fold only -- '_calibrate_and_tune ... never looks at the held-out video.' Comment:
'a bare LOVO mean at n?6 is not trustworthy on its own.'",
    "ROTATING ~10KB TELEMETRY BLACK BOX: run_telemetry.py keeps a tiny (~10,000-byte) rotating
JSONL runlog of execution velocity + machine health so a post-mortem on a kill with NO Python
traceback (OS kill, OOM, thermal/console-control event) is 'a bullseye instead of a guess.' A
fixed meta header line (label, started, pid, gpu_name, cpu_count) is ALWAYS kept; recent
snapshot/event lines form an on-disk ring trimmed to max_bytes; the file is rewritten ATOMICALLY
(tmp + os.replace) on every write so a hard kill never leaves it half-written. Caller-driven (no

```

background thread): snapshot(done,total) derives fps from the delta since the previous snapshot; event(kind,msg) records milestones WITH a health sample so the last line before a kill captures machine state at death.",

"TELEMETRY SIGNALS captured per sample: GPU via nvidia-smi -- temp_c, util_pct, power_w, current+max SM clock (sm_mhz/sm_max_mhz), vram_used_mb/vram_free_mb, plus a derived throttled flag (SM clock below 0.8x max, OR temp >= 84C, laptop Max-Q throttle territory). System via psutil -- cpu_pct, ram_used_gb/ram_total_gb/ram_pct. The docstring's worked example: 'it was at 26 fps, GPU 85C throttled to 1125/2100 MHz, 4.8 GB VRAM free, CPU 60%, RAM 21/32 GB -- then vanished.'",

"TELEMETRY GRACEFUL DEGRADATION + TESTABILITY: GPU stats absent nvidia-smi -> None; system stats absent psutil -> None; collection NEVER raises into the caller (_safe wraps collectors, _write swallows OSError) -- 'telemetry must never break the run it observes.' The clock and both collectors (gpu_fn, sys_fn) are injectable, so fps/rotation/parsing are unit-tested with no real GPU, no psutil, no wall-clock. It is the OBSERVABILITY companion to the ExecutionPolicy CONTROL layer (policy decides wait/abort/resume; telemetry records why a run slowed or died and lets execution velocity be compared across quality iterations).",

"DEFAULT-OFF DISCIPLINE: Every Variant knob defaults to control behaviour (no gate, no learned filter), so 'a variant that merely carries a new, disabled cue is byte-identical to control -- the core no-regression guarantee.' CONTROL = Variant(name='control') is the pinned/frozen baseline and 'rollback = flip the variant/config, never git revert.' New cues are recorded in cue_flags default-OFF; min_crossings/min_players gates = 0 (off); tune_threshold=False and calibrate=None default OFF -> unchanged behaviour. fusion_golden persists person features but they enter only as feature columns the learned scorer weights, 'no special-casing, default-OFF until the golden ?F1 justifies them.'",

"CI-GATED: .github/workflows/ci.yml runs 4 gates on every PR (concurrency-cancel, contents:read perms, Python 3.13). (1) Ruff lint -- pyflakes/pycodestyle + bugbear over the WHOLE repo (also the repo-wide invalid-syntax guard reaching scripts/ and training/, beyond the backend-only py_compile sweep). (2) Mypy lenient green baseline with an explicit per-module quarantine list (ratchet-down), typed core deps installed so real signatures are seen. (3) Syntax sweep + import smoke with NO GPU stack (no cv2/torch/numpy) -- guards the 7fbb0 class of breakage (a SyntaxError / module-level import error shipping while the suite stays green; test_import_smoke stubs absent deps). (4) Full offline mocked suite via run_all_tests.py with a coverage floor (--cov-fail-under=70), calibrated 2026-06-11 against the CI lane's measured 72% (local-with-obsreport 73%), ratchet UP never down without an owner decision."

],

"numbers": [

"Telemetry runlog cap: max_bytes default 10,000 bytes (~10 KB)",

"Throttle thresholds: SM clock < 0.8x max (_THROTTLE_CLOCK_FRAC=0.8) OR temp >= 84.0 C (_THROTTLE_TEMP_C)",

"nvidia-smi: 7 fields per query (temperature.gpu, utilization.gpu, power.draw, clocks.sm, clocks.max.sm, memory.used, memory.free); 4.0s timeout",

"Variant.spec_hash = sha1 truncated to 12 chars",

"Feature substrate: 5 shuttle features (distill_local.FEATURE_NAMES) + ~5 person features (fusion_features.PERSON_FEATURE_NAMES) per candidate window",

"fusion_golden defaults: frame_skip=3, iou=0.5, log_every_sec=15.0",

"compare/evaluate_variant default iou=0.5, honest=True, honest_stats=True; tune_threshold grid 0.05..0.95 (k=1..19) on the train fold",

"CI: 4 jobs, Python 3.13, coverage floor --cov-fail-under=70 (calibrated against measured 72% CI lane / 73% local)",

"Honest stats n?6 videos -> bootstrap CI + exact paired sign test (C1) required before trusting a LOVO mean"

],

"source_refs": [

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\utils\\run_telemetry.py (module docstring L1-30; gpu_stats L56-86; throttle consts L45-46; system_stats L89-104; RunTelemetry.snapshot/event L148-178; atomic _write L196-205; rotation _append L189-194; injectable clock/collectors L128-137, L207-216)",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\eval\\fusion_golden.py (docstring 'persist once / re-score forever' L1-16; SHUTTLE/PERSON feature names L32-35; extract_video JSON shape L49-120; single-pass detection L72-93; resumable + --smallest-first run() L123-167)",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\eval\\experiment.py (header detection-vs-decision thesis L1-37; Variant default-OFF L76-141; spec_hash L148-153; CONTROL L156-158; LOVO evaluate_variant L296-356; honest CI/sign-test/F1@k L359-401; compare label_set_hash L403-448; record_experiment leaderboard L504-540; load_video_candidates DB bridge

```

L543-561; DEFAULT_LEADERBOARD_PATH L63)",
    "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.github\\workflows\\ci.yml (lint
L20-31; typecheck L37-48; import-smoke L54-65; full-suite + coverage floor L67-92)"
  ]
}
]

```

TASK: write a self-contained Python script at scratch/quality_report.py that uses reportlab (Platypus) + matplotlib to generate a polished multi-page PDF at scratch/Rally_Detection_Quality_Report.pdf, then RUN it (python scratch/quality_report.py) and confirm the PDF exists and is non-trivial in size (ls -la). Iterate if it errors.

REPORT STRUCTURE (academic): Title + one-line subtitle ("Project report ? toward a paper") + abstract; 1 Introduction & Problem (rally detection; the local/cheap constraint; failure modes = held-shuttle false positives + over-segmentation); 2 System & Method (the 2-layer model: shuttle-trajectory substrate -> candidate windows -> learned scorer; signals as confidence-weighted feature columns, default-OFF, no special-casing, promote-only-on-lift); 3 Evaluation Methodology (THE HEART: the n=6 owned golden corpus; ROORA AS AN ORACLE LABEL GENERATOR ? the contract is with the golden labels treated as trustworthy ground truth, not their source, so rigor is decoupled from the vendor and the corpus scales by adding oracles; leave-one-video-out; bootstrap 95% CI + paired sign test; promote-only-on-lift; replayable feature substrate "persist once, re-score forever"); 4 Results (TABLES: Gemini/owned baselines clean vs multi-court; Gate-A PROMOTED measured win; the learned-variant fix; fusion P3 person + P4 audio with F1/?F1/CI/sign-test; PLUS a matplotlib bar chart of the fusion ablation saved as a PNG and embedded); 5 Related Work (honest: TAL/TAS + late fusion = standard prior art, "cite don't claim"; the novel combination = report-first reconciler-as-linter + the disciplined small-N promotion gate); 6 Limitations (n=6; CIs span 0 for person+audio; court polygons absent so person-cue counts spectators; served path mock-tested); 7 Planned Work (grow the corpus via the oracle; activate court polygons; audio-on-shuttle-directly; shuttle-dropped cue; owned model Qwen3-VL Apache-2.0; execution-policy resilience); 8 Engineering & Reproducibility (telemetry black box; CI gates; experiment leaderboard); Conclusion.

REPORTLAB PITFALLS (obey strictly): the built-in Helvetica font does NOT contain glyphs like ? ? ± × ? they render as black boxes. EITHER register a Unicode TTF (try scratch/arial.ttf if it exists via TTFont, or C:/Windows/Fonts/arial.ttf) and use it for all styles, OR write ASCII only ("Delta-F1", ">=", "+/-"). NEVER use Unicode subscripts/superscripts (use <sub>/<sup> tags in Paragraphs). Keep tables readable (reportlab Table with TableStyle, header shading, grid). Use a clean academic look (Times-like or the registered TTF; section headings; page numbers in a footer). Make it genuinely good ? this represents months of disciplined work.

Return: the absolute path to the generated PDF, its size, page count if known, and a 2-line note on anything you had to adapt.

2. assistant (2026-06-14T06:52:34.699Z)

I'll start by checking the environment, then write and run the script.

3. user (2026-06-14T06:52:37.752Z)

```

reportlab 4.5.1
matplotlib 3.10.9
scratch/arial.ttf
C:/Windows/Fonts/arial.ttf

```

4. assistant (2026-06-14T06:52:40.611Z)

Both libraries are available and both font paths exist. Now I'll write the script.

5. assistant (2026-06-14T06:55:44.426Z)

```

"""Generate a paper-style quality report PDF for the rally-detection system.

```

Self-contained: uses reportlab (Platypus) + matplotlib. Produces a polished multi-page PDF at scratch/Rally_Detection_Quality_Report.pdf. All numbers are

verified facts pulled from the codebase (docs/QUALITY_ITERATIONS.md and the eval_baselines/*.json leaderboard records); nothing here is invented.

Run: `python scratch/quality_report.py`
"""

```
import os
```

```
import matplotlib
```

```
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from reportlab.lib import colors
from reportlab.lib.enums import TA_CENTER, TA_JUSTIFY, TA_LEFT
from reportlab.lib.pagesizes import A4
from reportlab.lib.styles import ParagraphStyle, getSampleStyleSheet
from reportlab.lib.units import cm, mm
from reportlab.pdfbase import pdfmetrics
from reportlab.pdfbase.ttfonts import TTFont
from reportlab.platypus import (
    BaseDocTemplate,
    Frame,
    Image,
    PageBreak,
    PageTemplate,
    Paragraph,
    Spacer,
    Table,
    TableStyle,
)
```

```
HERE = os.path.dirname(os.path.abspath(__file__))
PDF_PATH = os.path.join(HERE, "Rally_Detection_Quality_Report.pdf")
CHART_PATH = os.path.join(HERE, "_fusion_ablation_chart.png")
```

Fonts: register a Unicode TTF so glyphs like the delta, $\geq$, $\pm$, $\times$ render.

```
FONT_REGULAR = "Helvetica"
FONT_BOLD = "Helvetica-Bold"
FONT_ITALIC = "Helvetica-Oblique"
```

```
_arial_candidates = [
    os.path.join(HERE, "arial.ttf"),
    r"C:/Windows/Fonts/arial.ttf",
]
_arialbd_candidates = [
    r"C:/Windows/Fonts/arialbd.ttf",
    os.path.join(HERE, "arialbd.ttf"),
]
_ariali_candidates = [
    r"C:/Windows/Fonts/ariali.ttf",
    os.path.join(HERE, "ariali.ttf"),
]
```

```
def _first_existing(paths):
    for p in paths:
        if os.path.exists(p):
            return p
    return None
```

```

_reg = _first_existing(_arial_candidates)
if _reg:
    try:
        pdfmetrics.registerFont(TTFont("UI", _reg))
        FONT_REGULAR = "UI"
        bd = _first_existing(_arialbd_candidates)
        it = _first_existing(_ariali_candidates)
        pdfmetrics.registerFont(TTFont("UI-Bold", bd if bd else _reg))
        pdfmetrics.registerFont(TTFont("UI-Italic", it if it else _reg))
        FONT_BOLD = "UI-Bold"
        FONT_ITALIC = "UI-Italic"
        # Map the family so <b>/<i> inline tags resolve to the TTF variants.
        pdfmetrics.registerFontFamily(
            "UI", normal="UI", bold="UI-Bold", italic="UI-Italic", boldItalic="UI-Bold"
        )
    except Exception as exc: # pragma: no cover - defensive
        print("Font registration fell back to Helvetica:", exc)
        FONT_REGULAR, FONT_BOLD, FONT_ITALIC = (
            "Helvetica",
            "Helvetica-Bold",
            "Helvetica-Oblique",
        )

UNICODE_OK = FONT_REGULAR == "UI"
DELTA = "?" if UNICODE_OK else "Delta" # Greek capital delta
GE = "?" if UNICODE_OK else ">="
PM = "±" if UNICODE_OK else "+/-"
TIMES = "×" if UNICODE_OK else "x"
ARROW = "?" if UNICODE_OK else "->"

```

Palette

```

NAVY = colors.HexColor("#13294B")
STEEL = colors.HexColor("#2C5F8A")
ACCENT = colors.HexColor("#C8102E")
LIGHT = colors.HexColor("#EAF0F6")
MIDGREY = colors.HexColor("#5A6675")
HEADER_BG = colors.HexColor("#1F4E6B")
ROW_ALT = colors.HexColor("#F2F6FA")
GREEN = colors.HexColor("#1B7A3D")

```

Styles

```

styles = getSampleStyleSheet()

```

```

def style(name, **kw):
    base = ParagraphStyle(name, parent=styles["Normal"], **kw)
    return base

```

```

BODY = style(
    "Body",
    fontName=FONT_REGULAR,
    fontSize=9.5,
    leading=14,
    alignment=TA_JUSTIFY,
    spaceAfter=6,
    textColor=colors.HexColor("#1A1A1A"),
)
BODY_TIGHT = style(
    "BodyTight",

```

```

        fontName=FONT_REGULAR,
        fontSize=9.5,
        leading=13.5,
        alignment=TA_JUSTIFY,
        spaceAfter=3,
    )
    H1 = style(
        "H1",
        fontName=FONT_BOLD,
        fontSize=14,
        leading=18,
        textColor=NAVY,
        spaceBefore=14,
        spaceAfter=6,
    )
    H2 = style(
        "H2",
        fontName=FONT_BOLD,
        fontSize=11,
        leading=15,
        textColor=STEEL,
        spaceBefore=8,
        spaceAfter=4,
    )
    ABSTRACT = style(
        "Abstract",
        fontName=FONT_ITALIC,
        fontSize=9.5,
        leading=14,
        alignment=TA_JUSTIFY,
        textColor=colors.HexColor("#222222"),
        leftIndent=10,
        rightIndent=10,
        spaceAfter=6,
    )
    CAPTION = style(
        "Caption",
        fontName=FONT_ITALIC,
        fontSize=8,
        leading=11,
        alignment=TA_CENTER,
        textColor=MIDGREY,
        spaceBefore=2,
        spaceAfter=8,
    )
    TITLE = style(
        "Title2",
        fontName=FONT_BOLD,
        fontSize=22,
        leading=27,
        alignment=TA_CENTER,
        textColor=NAVY,
        spaceAfter=4,
    )
    SUBTITLE = style(
        "Subtitle",
        fontName=FONT_ITALIC,
        fontSize=11.5,
        leading=15,
        alignment=TA_CENTER,
        textColor=ACCENT,
        spaceAfter=2,
    )
    AUTHORLINE = style(
        "Author",

```

```

        fontName=FONT_REGULAR,
        fontSize=9,
        leading=13,
        alignment=TA_CENTER,
        textColor=MIDGREY,
        spaceAfter=14,
    )
TBL_CELL = style("TblCell", fontName=FONT_REGULAR, fontSize=8.3, leading=10.5)
TBL_HEAD = style(
    "TblHead",
    fontName=FONT_BOLD,
    fontSize=8.4,
    leading=10.5,
    textColor=colors.white,
)
KEYBOX = style(
    "KeyBox",
    fontName=FONT_REGULAR,
    fontSize=9,
    leading=13,
    textColor=NAVY,
    leftIndent=6,
    rightIndent=6,
    spaceAfter=2,
)

def P(text, st=BODY):
    return Paragraph(text, st)

```

Matplotlib chart: fusion ablation (uses full-precision JSON floats)

```

def build_chart():
    # Full-precision values from eval_baselines/experiment_leaderboard.json
    labels = ["shuttle-only", "+person (P3)", "+person+audio (P4)"]
    f1 = [0.3068611057861412, 0.2863398846047612, 0.3655826552797436]
    # 95% bootstrap CI half-widths are only defined for the incremental
    # deltas, not for absolute F1; we annotate deltas as text instead.
    bar_colors = ["#2C5F8A", "#9AA7B5", "#1B7A3D"]

    fig, ax = plt.subplots(figsize=(6.6, 3.5), dpi=150)
    x = range(len(labels))
    bars = ax.bar(x, f1, color=bar_colors, width=0.58, edgecolor="#1A2A38", linewidth=0.6)

    for i, (b, v) in enumerate(zip(bars, f1)):
        ax.text(
            b.get_x() + b.get_width() / 2,
            v + 0.006,
            f"{v:.3f}",
            ha="center",
            va="bottom",
            fontsize=10,
            fontweight="bold",
            color="#13294B",
        )

    # delta annotations between consecutive bars
    ax.annotate(
        "?F1 = -0.021\n(CI spans 0)",
        xy=(0.5, 0.33),
        ha="center",

```



```

        fontsize=8,
        color="#C8102E",
    )
    ax.annotate(
        "?F1 = +0.079\n(CI spans 0)",
        xy=(1.5, 0.39),
        ha="center",
        fontsize=8,
        color="#1B7A3D",
    )

    # reference line: heuristic LOVO floor
    ax.axhline(0.480, color="#C8102E", linestyle="--", linewidth=1.0, alpha=0.8)
    ax.text(
        2.42, 0.487, "heuristic LOVO floor 0.480", ha="right", va="bottom",
        fontsize=7.5, color="#C8102E",
    )

    ax.set_xticks(list(x))
    ax.set_xticklabels(labels, fontsize=9)
    ax.set_ylabel("LOVO F1 (IoU >= 0.5, n=6)", fontsize=9.5)
    ax.set_ylim(0, 0.55)
    ax.set_title(
        "Fusion ablation: incremental cue contribution (honest LOVO)",
        fontsize=10.5,
        fontweight="bold",
        color="#13294B",
    )
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.yaxis.grid(True, linestyle=":", alpha=0.4)
    ax.set_axisbelow(True)
    fig.tight_layout()
    fig.savefig(CHART_PATH, bbox_inches="tight", facecolor="white")
    plt.close(fig)

```

Table helper

```

def make_table(data, col_widths, header_rows=1, align_centers=None, font_size=8.3):
    """data: list of rows; row[i] is either a str or a Paragraph."""
    wrapped = []
    for r_idx, row in enumerate(data):
        new_row = []
        for c_idx, cell in enumerate(row):
            if isinstance(cell, Paragraph):
                new_row.append(cell)
            else:
                st = TBL_HEAD if r_idx < header_rows else TBL_CELL
                new_row.append(Paragraph(str(cell), st))
        wrapped.append(new_row)

    t = Table(wrapped, colwidths=col_widths, repeatRows=header_rows)
    cmds = [
        ("BACKGROUND", (0, 0), (-1, header_rows - 1), HEADER_BG),
        ("TEXTCOLOR", (0, 0), (-1, header_rows - 1), colors.white),
        ("VALIGN", (0, 0), (-1, -1), "MIDDLE"),
        ("TOPPADDING", (0, 0), (-1, -1), 4),
        ("BOTOMPADDING", (0, 0), (-1, -1), 4),
        ("LEFTPADDING", (0, 0), (-1, -1), 5),
        ("RIGHTPADDING", (0, 0), (-1, -1), 5),
        ("GRID", (0, 0), (-1, -1), 0.4, colors.HexColor("#B8C4D0")),
    ]

```

```

        ("LINEBELOW", (0, header_rows - 1), (-1, header_rows - 1), 1.0, NAVY),
        ("ROWBACKGROUNDS", (0, header_rows), (-1, -1), [colors.white, ROW_ALT]),
    ]
    if align_centers:
        for c in align_centers:
            cmds.append(("ALIGN", (c, 0), (c, -1), "CENTER"))
    t.setStyle(TableStyle(cmds))
    return t

```

Page decoration: header rule + footer page number

```

def on_page(canvas, doc):
    canvas.saveState()
    w, h = A4
    # top rule
    canvas.setStrokeColor(NAVY)
    canvas.setLineWidth(0.8)
    canvas.line(2 * cm, h - 1.4 * cm, w - 2 * cm, h - 1.4 * cm)
    canvas.setFont(FONT_ITALIC, 7.5)
    canvas.setFillColor(MIDGREY)
    canvas.drawString(2 * cm, h - 1.25 * cm, "Local, Cheap, AI Rally Detection for Racket
Sports")
    canvas.drawRightString(
        w - 2 * cm, h - 1.25 * cm, "Project Quality Report ? toward a paper"
    )
    # bottom rule + page number
    canvas.setStrokeColor(colors.HexColor("#C6D0DA"))
    canvas.setLineWidth(0.6)
    canvas.line(2 * cm, 1.4 * cm, w - 2 * cm, 1.4 * cm)
    canvas.setFont(FONT_REGULAR, 8)
    canvas.setFillColor(MIDGREY)
    canvas.drawCentredString(w / 2.0, 1.0 * cm, f"{doc.page}")
    canvas.drawString(2 * cm, 1.0 * cm, "badminton-highlight-indexer")
    canvas.drawRightString(w - 2 * cm, 1.0 * cm, "Internal · honest measured results only")
    canvas.restoreState()

def on_first_page(canvas, doc):
    # Title page gets a colored banner instead of the running header.
    canvas.saveState()
    w, h = A4
    canvas.setFillColor(NAVY)
    canvas.rect(0, h - 4.4 * cm, w, 4.4 * cm, fill=1, stroke=0)
    canvas.setFillColor(ACCENT)
    canvas.rect(0, h - 4.5 * cm, w, 0.1 * cm, fill=1, stroke=0)
    canvas.restoreState()
    on_page(canvas, doc)

```

Build the story

```

def build_story():
    story = []

    # --- Title block (sits below the banner) ---
    story.append(Spacer(1, 0.2 * cm))
    story.append(Paragraph("Local, Cheap, AI Rally Detection<br/>for Racket Sports", TITLE))
    story.append(Paragraph("Project report &mdash; toward a paper", SUBTITLE))

```

```

story.append(
  Paragraph(
    "An honest empirical study of a shuttle-trajectory rally segmenter, "
    "its disciplined small-N promotion gate, and a fusion ablation",
    AUTHORLINE,
  )
)

# --- Abstract ---
story.append(Paragraph("Abstract", H1))
abstract = (
  "We present a fully local, low-cost rally-detection system for racket sports "
  "(badminton first) that segments a match video into rally <i>[start, end]</i> "
  "intervals. The system is built on a frozen shuttle-trajectory substrate (a WASB CNN "
  "that emits per-frame shuttle position), from which a velocity signal seeds "
  "recall-oriented candidate windows that a tiny pure-numpy logistic-regression scorer "
  "grades. Every additional cue enters as a confidence-weighted feature column that the "
  "scorer learns to weight &mdash; never as a hand-coded branch &mdash; and ships "
  "<b>default-OFF</b>, promoted to the served default only on a measured held-out lift. "
  "We evaluate against a small owned golden corpus of <b>n=6</b> labelled matches,
treating "
  "the labelling vendor (Roora) as an interchangeable <i>oracle</i>: the contract is with
the "
  "golden labels as trustable ground truth, not with their provenance, so evaluation rigor
"
  "is decoupled from the vendor and the corpus scales by adding oracles. Using leave-one-"
  "video-out cross-validation with bootstrap 95% confidence intervals and a paired sign "
  "test, our first fused cue &mdash; a net-crossing gate (Gate&nbsp;A) &mdash; is honestly
"
  f"<b>promoted</b> ({DELTA}F1 = +0.074, 6/6 matches, sign-test p = 0.031, CI [+0.042,
+0.104]), "
  "cutting over-segmentation from 1.68 to 1.01. Two further cues (a person cue, then
audio) "
  "do <i>not</i> clear the bar at n=6 (both CIs span 0) and stay default-OFF. We report
these "
  "negative results deliberately as the honest ablation backbone, frame the architecture "
  "against standard prior art (temporal action localization / segmentation + late fusion),
"
  "and isolate the genuinely novel piece: a report-first reconciler-as-linter paired with
a "
  "disciplined small-N promotion gate."
)
story.append(Paragraph(abstract, ABSTRACT))

# Key-result strip
kr = [
  [P(f"<b>Gemini wrapper (cloud ref.)</b>", TBL_CELL), P("clean 0.93 / multi-court 0.66",
TBL_CELL)],
  [P("<b>Gen-0 owned head (LOVO)</b>", TBL_CELL), P("clean 0.744 / multi-court 0.561",
TBL_CELL)],
  [P("<b>Gate-A (PROMOTED)</b>", TBL_CELL), P(f"{DELTA}F1 +0.074, p=0.031, CI
[+0.042,+0.104]", TBL_CELL)],
  [P("<b>Golden corpus</b>", TBL_CELL), P("n=6 matches (262 rallies), Proprietary, minors
excluded", TBL_CELL)],
]
kt = Table(kr, colWidths=[5.2 * cm, 11.3 * cm])
kt.setStyle(
  TableStyle(
    [
      ("BACKGROUND", (0, 0), (0, -1), LIGHT),
      ("BOX", (0, 0), (-1, -1), 0.6, STEEL),
      ("INNERGRID", (0, 0), (-1, -1), 0.4, colors.HexColor("#C6D0DA")),
      ("VALIGN", (0, 0), (-1, -1), "MIDDLE"),
      ("TOPPADDING", (0, 0), (-1, -1), 3),
      ("BOTTOPPADDING", (0, 0), (-1, -1), 3),
    ]
  )
)

```

```

        ("LEFTPADDING", (0, 0), (-1, -1), 6),
    ]
)
)
story.append(Spacer(1, 0.2 * cm))
story.append(kt)
story.append(Paragraph("Headline results at a glance (IoU 0.5 vs human golden labels).",
CAPTION))

# ===== 1. Introduction =====
story.append(Paragraph("1&nbsp;&nbsp;&nbsp;Introduction & Problem", H1))
story.append(
    P(
        "A rally is the unit of play in racket sports: the span from serve contact to the
point "
        "ending. Indexing a long match video into its rallies is the substrate for
everything a "
        "coach or player wants &mdash; highlight reels, &ldquo;show me the longest
rally,&rdquo; "
        "per-rally statistics. Cloud vision-language models can do this well on clean
footage, but "
        "they are expensive per video, non-deterministic, and (critically) <b>cannot be a
training "
        "data source</b> for an owned model under their terms. Our constraint is the
opposite end of "
        "the design space: a system that runs <b>locally, cheaply, and reproducibly</b> on a
single "
        "consumer GPU, with paid LLMs allowed only as an optional inference fallback &mdash;
never "
        "in the training loop."
    )
)
story.append(
    P(
        "Two failure modes dominate and motivate the entire method. <b>(1) Held-shuttle
false "
        "positives:</b> a player holding or flicking the shuttle between points produces "
        "player-motion that the old velocity heuristic counted as a rally &mdash; the owner-
observed "
        "precision bug. <b>(2) Over-segmentation:</b> a single true rally gets shredded into
several "
        "short fragments. Crucially, the standard temporal-IoU metric is nearly <i>blind</i>
to "
        "over-segmentation, so a method can look fine on IoU while destroying rally
structure. Both "
        "failure modes are about the boundary between play and non-play, which is exactly
what the "
        "fusion gates in this report target."
    )
)

# ===== 2. System & Method =====
story.append(Paragraph("2&nbsp;&nbsp;&nbsp;System & Method", H1))
story.append(Paragraph("2.1&nbsp;&nbsp;&nbsp;A two-layer mental model", H2))
story.append(
    P(
        "<b>Layer 1 &mdash; Detection (&ldquo;where is the shuttle this frame?&rdquo;).</b>
A frozen "
        "WASB CNN emits a per-frame (x, y, visible) shuttle track. The shuttle is chosen as
the "
        "camera-<i>invariant</i> primary signal because &ldquo;shuttle in flight&rdquo; is a
near-"
        "synonym for &ldquo;a rally is happening,&rdquo; unlike player motion which fooled
the old "
        "heuristic on broadcast pan cameras. <b>Layer 2 &mdash; Segmentation (&ldquo;when is

```

```

a "
    "rally?">).</b> From the per-frame track we compute a shuttle <i>velocity</i>
signal, "
    "which seeds recall-oriented candidate windows; a learned scorer then grades each
window "
    f"into final [start, end] rallies measured by temporal IoU. The chain is: frames
{ARROW} "
    f"WASB frozen trajectory {ARROW} velocity {ARROW} shuttle-seeded candidate windows
{ARROW} "
    "learned scorer weights per-window feature columns {0} graded rallies. The original
v0 "
    "segmentation was three hand-tuned knobs (velocity threshold, merge gap, minimum
window "
    "duration); the learned scorer replaces that hand-tuning.".format(ARROW)
    )
    )
    story.append(
        P(
            "A key honesty property follows from the data: golden labels are rally <i>times</i>
"
            "(when), not (x, y) <i>coordinates</i> (where). They therefore cannot retrain the
frozen "
            "WASB detector &mdash; they can only grade and tune the segmentation layer. The
discipline "
            "is &ldquo;calibrate on some rallies, measure on held-out ones.&rdquo;")
        )
    )
    story.append(Paragraph("2.2&nbsp;&nbsp;&nbsp;Signals as feature columns, not branches", H2))
    story.append(
        P(
            "Every producer (detector / analyzer / reconciler) emits a <b>signal = (value, "
            "confidence)</b> persisted to a namespaced column <font face=\"{0}\">"
            "&lt;producer&gt;__&lt;key&gt;</font>. The scoring model consumes those columns and
"
            "<b>learns to weight</b> them; there are <b>no hand-coded if/else branches</b> in
the "
            "decision path. Producers propose; the scorer arbitrates; the experiment harness and
the "
            "golden set decide what is promoted. The scorer itself is deliberately a tiny pure-
numpy "
            "L2 logistic regression (ADR-004), <i>not</i> a neural net, so it survives small-N.
A cue "
            "that does not earn its keep simply stays at weight ~0 &mdash; it costs nothing
because it "
            "is an unused column, not a live branch.".format(FONT_ITALIC)
        )
    )
    story.append(
        P(
            "Fusion has two realizations. The <b>decision-level gate</b> (label-free, ships
first) "
            f"composes Gate&nbsp;A (net-crossings {GE} 2 &mdash; the structural held-shuttle /
service-"
            "feed fix) and Gate&nbsp;B (a person cue). The <b>feature-level</b> path (the
primary, "
            "learned route) lets cues emit a small set of scale/translation-<i>invariant</i>
per-window "
            "features (counts, ratios, zone-membership, two-sidedness &mdash; never raw
coordinates, "
            "which would memorize this court and camera) that join the logistic feature vector.
The "
            "shuttle stays primary; person is a gate-plus-feature, never the sole trigger, and
single-"
            "cue segmenters remain registered fallbacks."
        )
    )

```

```

)

# ===== 3. Evaluation Methodology (the heart) =====
story.append(Paragraph("3   Evaluation Methodology", H1))
story.append(
    P(
        "This section is the heart of the work: the system's credibility rests on how
it is "
        "measured, not on any single number."
    )
)
story.append(Paragraph("3.1   The owned golden corpus and the oracle contract",
H2))
story.append(
    P(
        "Evaluation uses a small owned golden corpus of n=6 labelled matches "
        "(<font face=\"{0}\">adarsh_avi_singles, kushagra_singles, largetest_doubles,
gbaaddy, "
        "testlarge_short, mahadevpura_singles</font>; 262 rallies total), grown by the
owner, "
        "classified Proprietary, with minors excluded entirely and no biometric/identity
labelling. "
        "The labelling vendor, Roora, is treated as an interchangeable oracle
label "
        "generator</b>: the system's contract is with the golden labels &mdash;
assumed- "
        "trustable ground truth, validated by automated checks plus a ~20% human spot-check
at a "
        f"{PM}0.3 s boundary tolerance &mdash; not with their provenance.
Evaluation "
        "rigor (interval matching, LOVO, the ship-gate) is therefore decoupled from
the "
        "vendor, and the corpus scales by adding or swapping oracles; any oracle's
labels "
        "import directly in the same schema. This is the project's moat: a consented dataset
plus an "
        "honest eval harness, not the weights.".format(FONT_ITALIC)
    )
)
story.append(Paragraph("3.2   Leave-one-video-out, with honest statistics", H2))
story.append(
    P(
        "Learned variants are scored leave-one-video-out (LOVO), grouped by match:
train on "
        "the other videos' candidates, predict the held-out video, score against its
labels. "
        "We never hold out a window from the same video &mdash; windows within a
video are "
        "correlated and would flatter the score (the field name is leave-one-group/subject-
out CV). "
        "Every tunable knob &mdash; the operating-point threshold, the calibrator &mdash; is
fit on "
        "the training fold only and never looks at the held-out video. Because a bare
LOVO "
        f"mean at n {TIMES if False else GE} 6 is not trustworthy on its own, we attach
<b>honest "
        "statistics</b>: a bootstrap 95% confidence interval per metric, a paired "
        f"{DELTA}F1 CI, and an exact paired sign test. The promotion rule is encoded
directly: "
        f"a lift is real only when the {DELTA}F1 CI clears 0 and the sign test
agrees. This is "
        "necessary because the sign test needs sample size &mdash; at n=6 a ~+0.08 lift
cannot reach "
        "p < 0.05; roughly n=10 with 9 wins gives p ≈ 0.011."
    )
)

```

```

)
story.append(Paragraph("3.3   Persist once, re-score forever", H2))
story.append(
    P(
        "The expensive, non-deterministic per-frame <b>detection</b> (shuttle WASB
trajectory, person "
        "tracks, audio hits) runs <b>once</b> per video on the GPU/WSL and is persisted as a
small "
        "replayable feature substrate (one JSON per video of resolution-invariant shuttle +
person "
        "features per candidate window). The cheap, deterministic <b>decision</b>
(windowing, gates, "
        "the logistic scorer) then re-scores any number of variants <b>offline</b>, with no
GPU and "
        "zero production risk. A <i>Variant</i> is a declarative re-scoring recipe, not a
code branch; "
        "given persisted features it produces the same intervals deterministically. This is
what makes "
        "every experiment a zero-risk offline re-score &mdash; the reproducibility thesis of
the whole "
        "harness."
    )
)

story.append(PageBreak())

# ===== 4. Results =====
story.append(Paragraph("4   Results", H1))
story.append(Paragraph("4.1   Baselines that drive strategy", H2))
story.append(
    P(
        "Table 1 anchors strategy. Two reference footages are reported:
<b>Mahadevpura</b> "
        "(clean-ish, single court) and <b>Kushagra</b> (multi-court, the hard case, contrast
"
        f"1.4{TIMES}). The reading is blunt: clean footage is commoditized (a Gemini serve
reaches "
        "0.93), but multi-court drops the cloud wrapper to <b>0.66</b> &mdash; and that 0.66
is the "
        "owned model's ship target. The owned Gen-0 head already reaches 0.561 multi-court
at n=6."
    )
)

tbl1 = [
    ["Method", "Mahadevpura (clean)", "Kushagra (multi-court)"],
    ["Heuristic (velocity windowing)", "0.657", "0.157"],
    ["Gen-0 owned head (LOVO, n=6)", "0.744", "0.561"],
    ["Two-stage WASB + Gemini refine", "0.83", "&mdash; (not measured)"],
    ["Gemini wrapper (gemini-flash-latest)", "0.93", "0.66"],
]
story.append(make_table(tbl1, [7.2 * cm, 4.65 * cm, 4.65 * cm], align_centers=[1, 2]))
story.append(
    Paragraph(
        "Table 1. Rally-detection F1 at IoU 0.5 vs human golden labels. "
        "Clean footage is commoditized; multi-court 0.66 is the owned-model ship target.",
        CAPTION,
    )
)

story.append(
    P(
        "Per-video detail on the cloud reference (gemini-flash-latest): on Mahadevpura, F1
0.93, "
        "precision 1.00, recall 0.87, mean IoU 0.80, boundary error 0.7 s (n<sub>gt 31,
n_pred 27; "
        "3 of 4 false negatives were detected by Gemini but rejected by a 3.0 s

```

```

minimum-segment "
    "filter on 2.5&nbsp;s rallies). On the harder Kushagra, F1 0.66, precision 0.69,
recall 0.62, "
    "mean IoU 0.73, boundary error 1.2&nbsp;s (n_gt 32, n_pred 29)."
)
)

story.append(Paragraph("4.2&nbsp;&nbsp;Gate-A: the first honestly-promoted cue", H2))
story.append(
    P(
        f"Gate&nbsp;A (net-crossings {GE} 2) is the first cue promoted through the harness,
because it "
        "is data-backed: real rallies on multi-court matches cross the net 2 or more times
<b>97% of "
        "the time</b> (median 5), whereas a single service feed crosses about once. The
measured "
        "result on the recall-oriented proposal candidate set is unambiguous
(Table&nbsp;2)."
    )
)
tbl2 = [
    ["Metric", "Result"],
    [f"{DELTA}F1 (Gate-A vs control)", "<b>+0.074</b>"],
    ["Golden matches won", "6 / 6"],
    ["Paired sign-test p", "<b>0.031</b>"],
    [f"95% CI on {DELTA}F1", "[+0.042, +0.104] (clears 0)"],
    ["Over-segmentation (segment-count ratio)", f"1.68 {ARROW} 1.01"],
    ["Recommended config", "<font face=\"%s\">indexing.fusion.min_crossings = 2</font>" %
FONT_ITALIC],
]
story.append(make_table(tbl2, [8.3 * cm, 8.2 * cm], align_centers=[1]))
story.append(
    Paragraph(
        "Table 2. Gate-A promotion statistics (2026-06-14). The CI clears 0 and the sign
test agrees "
        "? the only fusion cue to date that does.",
        CAPTION,
    )
)
story.append(
    P(
        "Real-footage corroboration (no golden labels, so this is a divergence check, not an
F1): an "
        "offline re-score of GX010125_1080p's cached trajectory took 45 candidates from
CONTROL's 45 "
        "rallies down to Gate-A's 27, dropping 18 low-crossing windows (~1.7&nbsp;min) of
likely "
        "held-shuttle / feed false positives. Caveat: this offline win is on the recall-
oriented "
        "proposal set; flipping the live served default remains owner-gated, since the
production "
        "fusion segmenter is still mock-tested rather than verified on real video."
    )
)

story.append(Paragraph("4.3&nbsp;&nbsp;The learned-variant fix (PR #130)", H2))
story.append(
    P(
        "An early end-to-end A/B of a window-level 5-feature logistic variant <i>lost</i> to
control "
        f"({DELTA}F1 = &minus;0.081, 2W/4L) and zeroed the testlarge video &mdash; but this
was a "
        "fixed-0.5-cutoff artifact, not a real defect of the stronger per-frame prototype.
PR&nbsp;#130 "
        "added an additive, default-OFF threshold-tuning step (operating point chosen on the

```



```

train "
    "fold) plus probability calibration (Platt / isotonic). Validated on the n=6 golden
set, mean "
    "F1 rose <b>0.307 &arr; 0.423 (+0.116)</b>, testlarge recovered from 0.000 to
0.300, and the "
    "learned variant now beats CONTROL (0.388). The lesson &mdash; calibrate and tune
the "
    "operating point <i>inside</i> the LOVO loop &mdash; is now baked into the harness."
)
)

story.append(Paragraph("4.4&nbsp;&nbsp;&nbsp;Fusion ablation: person (P3) and audio (P4)", H2))
story.append(
    P(
        "Two further cues were measured incrementally on the same n=6 corpus (honest LOVO,
IoU 0.5). "
        "Neither clears the promotion bar &mdash; both CIs span 0 and both sign tests are
non-"
        "significant &mdash; so both stay <b>default-OFF</b>. We report them in full because
negative "
        "and uncertain results are the honest ablation backbone (Table&nbsp;3,
Figure&nbsp;1). Values "
        "below use the full-precision leaderboard floats; the doc rounds them."
    )
)
tbl3 = [
    ["Step", "F1", f"{DELTA}F1 vs prior", "95% CI", "Sign test", "Clears 0?"],
    ["shuttle-only", "0.3069", "&mdash;", "&mdash;", "&mdash;", "&mdash;"],
    ["+person (P3)", "0.2863", "&minus;0.0205 (&minus;6.7%)", "[&minus;0.165, +0.162]",
"2W/4L, p=0.6875", "No"],
    ["+person+audio (P4)", "0.3656", "+0.0792 (+27.7%)", "[&minus;0.054, +0.203]", "4W/2L,
p=0.6875", "No"],
]
story.append(
    make_table(
        tbl3,
        [3.0 * cm, 1.7 * cm, 3.2 * cm, 3.0 * cm, 2.7 * cm, 1.9 * cm],
        align_centers=[1, 2, 3, 4, 5],
        font_size=7.6,
    )
)
story.append(
    Paragraph(
        "Table 3. Fusion ablation (n=6, honest LOVO, IoU 0.5; full-precision floats from "
        "experiment_leaderboard.json). Both records: IoU 0.5, label_set_hash 50d98ffbc370,
n=6.",
        CAPTION,
    )
)
story.append(
    P(
        "Person adds noise: without court polygons, the players-in-court count includes
spectators, "
        "so the cue is a wash (per-video: testlarge +0.40, gbaaddy &minus;0.22). Audio is
the more "
        "promising lead &mdash; it wins 4/6 (largetest +0.27, mahadevpura +0.27) and lifts
F1 to "
        "0.366, above the 0.307 shuttle-only baseline &mdash; but n=6 simply cannot confirm
it. The "
        "discipline holds: promote only on a lift whose CI clears 0."
    )
)
# Figure 1
if os.path.exists(CHART_PATH):
    img = Image(CHART_PATH, width=15.2 * cm, height=8.06 * cm)

```

```

story.append(Spacer(1, 0.1 * cm))
story.append(img)
story.append(
    Paragraph(
        "Figure 1. Incremental fusion ablation on the n=6 golden corpus. Neither cue's "
        "?F1 confidence interval clears zero; both remain default-OFF. Dashed line: the "
        "heuristic LOVO floor (0.480) the learned scorer must beat.",
        CAPTION,
    )
)

# ===== 5. Related Work =====
story.append(Paragraph("5   Related Work (cite, don't claim)", H1))
story.append(
    P(
        "A literature review (multi-agent web search, every citation adversarially verified "
        "&mdash; "
        "zero fabricated papers across ~40 checks) settles what is standard versus novel. "
        "The "
        "architecture's <b>skeleton is standard and must be cited, not claimed</b>. "
        "Candidate-windows-then-learned-scorer is <i>temporal action localization</i> (TAL, "
        "proposal-then-score). The reconciler's over-segmentation fixes are <i>temporal "
        "action "
        "segmentation</i> (TAS) refinement / post-processing. &ldquo;Signal not verdict / a "
        "learned "
        "arbiter over cue columns&rdquo; is late / score-level fusion realized as stacked "
        "generalization &mdash; roughly 20 years old and already present in racket sports "
        "(CVIU 2009; "
        "Ghosh et al. 2018 SVM-over-features). In-domain prior art (See et al. 2024/25 rally "
        "start/end; "
        "MonoTrack; Hsu et al. 2024; Goh et al. 2023 service fault; ShuttleSet) hard-codes "
        "the same "
        "cues as if/else verdicts."
    )
)
story.append(
    P(
        "Against that backdrop, <b>what is genuinely novel</b> is narrow and we park our "
        "claims there: "
        "(1) the <b>report-first reconciler-as-linter</b> bundle &mdash; contradiction- "
        "detection over a "
        "persisted temporal decision, with auditable structured interval corrections, per- "
        "rule A/B "
        "lift-gating <i>before</i> auto-apply, idempotency and fixed precedence, human-in- "
        "the-loop, and "
        "feeding a (value, confidence) scorer &mdash; for which no surveyed paper bundles "
        "all parts; and "
        "(2) this <b>disciplined small-N promotion gate</b> on a deliberately tiny golden "
        "set. "
        "Everything else (the fusion mechanism, the persist-once substrate, cross-check-and- "
        "correct, "
        "the no-regression harness) is textbook and we claim only the specific discipline, "
        "not the idea."
    )
)

# ===== 6. Limitations =====
story.append(Paragraph("6   Limitations", H1))
lims = [
    ("Sample size n=6.", "Enough for direction-checks and LOVO sign-tests, not enough to "
    "claim a "
    "general model. At n=6 a ~+0.08 lift cannot reach p < 0.05; corpus growth is the "
    "highest-ROI ")

```

```

        "unblock."),
        ("CIs span 0 for person and audio.", "Both fusion cues (P3, P4) have ?F1 confidence "
        "intervals straddling zero and non-significant sign tests, so neither is promotable yet
despite "
        "audio's encouraging +27.7% point estimate."),
        ("Court polygons absent.", "Without per-video court masks the person cue counts
spectators, "
        "which is the mechanical reason +person is a wash. Activating court polygons is the
obvious next "
        "fix."),
        ("Served path is mock-tested.", "The production fusion segmenter is verified offline on
cached "
        "trajectories, not yet on live real video; flipping the served default stays owner-
gated."),
        ("Detector domain gap unmeasured.", "WASB/TrackNet were trained on other rigs; detector
accuracy "
        "on our footage is not yet quantified, and every downstream cue inherits that gap."),
    ]
    for head, txt in lms:
        story.append(P(f"<b>{head}</b> {txt}", BODY_TIGHT))

# ===== 7. Planned Work =====
story.append(Paragraph("7  Planned Work", H1))
plans = [
    ("Grow the corpus via the oracle.", "The #1 lever. Priority: multi-court badminton (the
target "
    "distribution and where the ship target lives), then >=3 matches per new sport (table
tennis "
    "first &mdash; WASB transfers at F1 0.909). At n=10 with 9 wins, p ≈ 0.011
&mdash; "
    "significance is reachable within a month, adults only."),
    ("Activate court polygons.", "The intake manifest already collects court_corners,
net_line, and "
    "singles/doubles per video, so the detector can ignore people outside the court and
require >=2 "
    "players on court &mdash; the direct fix for the person-cue noise above."),
    ("Measure audio on shuttle-only directly.", "Audio is parked (E1 probe found ~2.2x "
    "discrimination, AUC ~0.6), not in reserve; the planned Fusion-P4 path lands it flag-
gated and "
    "default-OFF, promoted only on a measured LOVO lift."),
    ("Shuttle-dropped cue + wire Gate-A to production.", "The cheapest current win is wiring
the "
    "already-built net-crossing Gate-A into the live segmenter (it is presently eval-only).
A faithful "
    "serve-contact boundary definition structurally fixes the held-shuttle false positive;
a future "
    "&ldquo;shuttle-on-floor&rdquo; cue extends the same substrate reasoning."),
    ("Owned model: Qwen3-VL (Apache-2.0).", "The roadmap starts with the TAL head, not the
VLM, "
    "because boundary accuracy is the product and fine-tuned VLMs miss strict boundaries.
Base model "
    "Qwen3-VL (Apache-2.0), InternVL3 (MIT) fallback; commercial-clean licensing for code,
data, and "
    "weights; never train on paid-LLM outputs."),
    ("Execution-policy resilience.", "Motivated by a real Gemini hang with no client-side
timeout, a "
    "per-job declarative ExecutionPolicy (op_timeout_sec, max_wait_sec, self-scheduling via
the DB as "
    "coordinator) makes every long operation declare how it waits, aborts, and resumes."),
    ]
    for head, txt in plans:
        story.append(P(f"<b>{head}</b> {txt}", BODY_TIGHT))

# ===== 8. Engineering & Reproducibility =====
story.append(Paragraph("8  Engineering & Reproducibility", H1))

```

```

story.append(
    P(
        "<b>Telemetry black box.</b> A rotating ~10&nbsp;KB JSONL runlog (atomic tmp +
os.replace on "
        "every write, so a hard kill never leaves it half-written) records execution
velocity plus "
        "machine health, so a post-mortem on an OS/OOM/thermal kill with no Python traceback
is a "
        "bullseye instead of a guess. Each sample captures GPU temp/util/power/clocks/VRAM
(with a "
        "derived throttle flag at SM clock &lt; 0.8&times; max or temp &ge; 84&nbsp;C) and
CPU/RAM. "
        "It is the observability companion to the ExecutionPolicy control layer."
    )
)
story.append(
    P(
        "<b>Experiment leaderboard.</b> Each compare() appends one compact reproducible row
to "
        "<font face=\"%s\">eval_baselines/experiment_leaderboard.json</font> &mdash; a
tracked location "
        "that survives a clean checkout. Every row is pinned to (exact recipe via a 12-char
"
        "spec_hash, exact label set via a label_set_hash, exact IoU), so any regression is
traceable "
        "to a recipe change rather than drift." % FONT_ITALIC
    )
)
story.append(
    P(
        "<b>CI gates.</b> Four gates run on every PR (Python 3.13): Ruff lint over the whole
repo; a "
        "lenient-but-green mypy baseline with an explicit ratchet-down quarantine; a syntax
+ import "
        "smoke with no GPU stack (guarding the class of breakage where a module-level import
error "
        "ships while the suite stays green); and the full offline mocked suite with a
coverage floor "
        "(&minus;&minus;cov-fail-under=70, calibrated against the measured 72%). Combined
with default-"
        "OFF cues and a pinned CONTROL variant, this makes rollback a config flip, never a
git revert."
    )
)

# ===== Conclusion =====
story.append(Paragraph("9&nbsp;&nbsp;&nbsp;Conclusion", H1))
story.append(
    P(
        "The contribution of this work is less any single F1 number and more a <b>method for
being "
        "honest at small scale</b>. On a 6-match owned golden corpus, a shuttle-trajectory
substrate "
        "feeding a tiny logistic scorer, with every cue entering as a default-OFF feature
column and "
        "promoted only on a held-out lift whose confidence interval clears zero, produced
exactly one "
        f"honestly-promoted cue ({DELTA}F1 +0.074, p=0.031) and two cues that &mdash;
reported in full "
        "&mdash; do not yet clear the bar. By treating the labelling vendor as a swappable
oracle, the "
        "evaluation rigor is decoupled from data provenance, so the cheapest path to a
publishable "
        "result is simply more labelled matches: at n=10 with 9 wins, significance is one
month away. "
    )
)

```

```

novelty "
    "The architecture is deliberately standard where the literature is mature and claims
the honest "
    "only on the report-first reconciler and the small-N promotion discipline &mdash;
    "ablation backbone this report is built to be."
)
)

return story

def main():
    build_chart()

    doc = BaseDocTemplate(
        PDF_PATH,
        pagesize=A4,
        leftMargin=2 * cm,
        rightMargin=2 * cm,
        topMargin=1.8 * cm,
        bottomMargin=1.8 * cm,
        title="Local, Cheap, AI Rally Detection for Racket Sports - Quality Report",
        author="badminton-highlight-indexer",
    )
    frame = Frame(
        doc.leftMargin,
        doc.bottomMargin,
        doc.width,
        doc.height,
        id="body",
        leftPadding=0,
        rightPadding=0,
        topPadding=0,
        bottomPadding=0,
    )
    first = PageTemplate(id="First", frames=[frame], onPage=on_first_page)
    later = PageTemplate(id="Later", frames=[frame], onPage=on_page)
    doc.addPageTemplates([first, later])

    story = build_story()
    # After the first page, switch to the running-header template.
    story.insert(0, Spacer(1, 1.0 * cm)) # push title below the banner

    doc.build(story)

    size = os.path.getsize(PDF_PATH)
    print(f"Wrote {PDF_PATH} ({size} bytes)")
    # cleanup the temp chart? keep it for inspection
    return size

if __name__ == "__main__":
    main()

```

6. user (2026-06-14T06:55:44.450Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py (file state is current in your context ? no need to Read it back)

7. assistant (2026-06-14T06:55:47.586Z)

I need to fix one issue: the page template switch from First to Later won't happen automatically. Let me handle that with a frame break / NextPageTemplate. Let me run first to see what happens.

8. user (2026-06-14T06:55:49.677Z)

Wrote C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\Rally_Detection_Quality_Report.pdf (176532 bytes)

9. assistant (2026-06-14T06:55:53.491Z)

It ran. But I need to ensure the running header appears on pages 2+ (right now all pages use the First template's `on\_first\_page` which draws the banner on every page since I never switch templates). Let me verify the page setup and add a `NextPageTemplate` so only page 1 gets the banner. Let me check the current behavior and page count.

10. user (2026-06-14T06:55:55.793Z)

pages 7

11. assistant (2026-06-14T06:55:59.372Z)

7 pages. Now the issue: without a `NextPageTemplate`, BaseDocTemplate keeps using the first PageTemplate ("First") for ALL pages, so the navy banner is drawn on every page and the running header never appears. Let me fix that by adding `NextPageTemplate("Later")` at the start of the story.

12. user (2026-06-14T06:56:01.571Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py has been updated successfully. (file state is current in your context ? no need to Read it back)

13. assistant (2026-06-14T06:56:07.819Z)

Now I need to make page 1 use "First" and page 2+ use "Later". The story already starts with a `Spacer`. I'll set the first page template explicitly and emit `NextPageTemplate("Later")` so the second page onward switches. The cleanest way: add `NextPageTemplate("Later")` as the very first flowable (it takes effect at the next page break).

14. user (2026-06-14T06:56:10.558Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py has been updated successfully. (file state is current in your context ? no need to Read it back)

15. assistant (2026-06-14T06:56:15.957Z)

Now there is one more concern: the `on\_first\_page` calls `on\_page` which draws the running top rule + header text, AND draws the banner. On page 1 the banner overlaps the header text region (banner is the top 4.4cm, header text is at h-1.25cm which is inside the banner). That means on page 1 the header text "Local, Cheap..." would be drawn in dark grey on the navy banner ? barely visible. Let me make `on\_first\_page` NOT draw the top running header (only the bottom footer), so the banner is clean.

16. user (2026-06-14T06:56:21.740Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py has been updated successfully. (file state is current in your context ? no need to Read it back)

17. assistant (2026-06-14T06:56:24.571Z)

Now I need to refactor `on\_page` to split the footer into a `\_footer` helper that `on\_first\_page` can reuse.

18. user (2026-06-14T06:56:34.337Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py has been updated successfully. (file state is current in your context ? no need to Read it back)

19. assistant (2026-06-14T06:56:43.203Z)

Now the title page has the banner with a white caption, the title text below it, and a clean footer. One concern: with the inserted ``Spacer(1, 1.0cm)`` pushing content down plus the banner at top 4.4cm ? the title "Local, Cheap..." starts at the top margin (1.8cm) + spacer, which is ~2.8cm from top ? still inside the 4.4cm banner. The title is navy text and would sit ON the navy banner, invisible. I need to push the title fully below the banner. Banner bottom is at h-4.4cm; top margin is 1.8cm. So I need the spacer to push content to start below 4.4cm from top, i.e. spacer ? $4.4 - 1.8 + \text{a little} = \sim 3.0\text{cm}$. Let me increase the spacer.

20. user (2026-06-14T06:56:45.928Z)

The file `C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py` has been updated successfully. (file state is current in your context ? no need to Read it back)

21. assistant (2026-06-14T06:56:51.774Z)

Also the existing ``Spacer(1, 0.2*cm)`` at the start of ``build_story`` adds a bit more. Let me rerun and then render page 1 and a middle page to a PNG to visually verify there are no black-box glyphs and the banner/title look right.

22. user (2026-06-14T06:56:55.194Z)

Wrote `C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\Rally_Detection_Quality_Report.pdf` (176247 bytes)
pages 7

23. assistant (2026-06-14T06:56:58.239Z)

Let me render pages to images to visually confirm there are no black-box glyphs and the layout is clean. I'll try `pdftoppm`, then fall back to `PyMuPDF`.

24. user (2026-06-14T06:56:59.794Z)

`pymupdf` `PyMuPDF 1.27.2.3: Python bindings for the MuPDF 1.27.2 library.`

25. assistant (2026-06-14T06:57:02.729Z)

`PyMuPDF` is available. Let me render pages 1, 2, and 4 (results page with the table and chart) to PNGs so I can visually inspect for glyph boxes and layout.

26. user (2026-06-14T06:57:05.488Z)

saved `scratch/_preview_p1.png` 1191 1684
saved `scratch/_preview_p2.png` 1191 1684
saved `scratch/_preview_p4.png` 1191 1684
saved `scratch/_preview_p5.png` 1191 1684

27. assistant (2026-06-14T06:57:11.919Z)

Page 1 looks clean ? title sits below the banner, banner caption is white and readable, abstract renders, the key-result strip and ? glyph render correctly (no black boxes). Let me check the results page with the table and chart (page 4) and page 5.

28. assistant (2026-06-14T06:57:16.120Z)

Page 4 looks excellent ? running header is now present (top), Tables 1 and 2 render with header shading, grid, and alternating rows; ? glyphs render correctly. Let me check page 5 (the fusion ablation table + chart).

29. assistant (2026-06-14T06:57:23.536Z)

Page 5 is great ? Table 3 (fusion ablation) renders with the minus signs and ? glyphs correct,

and the matplotlib bar chart (Figure 1) is embedded cleanly with the heuristic-floor dashed line, $F1$ annotations, and value labels. Everything looks polished.

Let me verify page 2 quickly to confirm the running header switched correctly there too.

30. assistant (2026-06-14T06:57:28.813Z)

Page 2 confirms: running header at top, footer with page number at bottom, all arrows (?) and glyphs render correctly. The ``?`` in the chain renders fine. Everything is clean.

Now let me clean up the preview PNGs (they were just for inspection) and confirm the final PDF with ``ls -la``.

31. user (2026-06-14T06:57:30.221Z)

```
-rw-r--r-- 1 avidu 197609 176247 Jun 14 12:26 scratch/Rally_Detection_Quality_Report.pdf
-rw-r--r-- 1 avidu 197609 46211 Jun 14 12:26 scratch/_fusion_ablation_chart.png
-rw-r--r-- 1 avidu 197609 43356 Jun 14 12:26 scratch/quality_report.py
```

32. assistant (2026-06-14T06:57:43.366Z)

The PDF generated successfully and is verified visually across multiple pages.

```
**Generated PDF:** `C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\Rally_Detection_Quality_Report.pdf`
**Size:** 176,247 bytes (~172 KB) ? non-trivial
**Pages:** 7 (verified via pypdf)
**Script:** `C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py`
(runs clean: `python scratch/quality_report.py`)
```

Structure delivered exactly as specified: banner title page + subtitle ("Project report ? toward a paper") + abstract + headline-results strip; §1 Introduction (local/cheap constraint, held-shuttle FPs + over-segmentation, tIoU-blindness); §2 System & Method (2-layer model, signals-as-feature-columns/default-OFF/no-special-casing/promote-only-on-lift); §3 Evaluation Methodology (n=6 owned corpus, Roora-as-oracle contract, LOVO, bootstrap CI + paired sign test, persist-once/re-score-forever); §4 Results (3 tables ? baselines clean vs multi-court, Gate-A promotion, fusion P3/P4 ablation with $F1$ / $F1$ /CI/sign-test ? plus the embedded matplotlib Figure 1 bar chart with the 0.480 heuristic floor line); §5 Related Work (cite-don't-claim, TAL/TAS + late fusion standard, novel = report-first reconciler + small-N gate); §6 Limitations; §7 Planned Work; §8 Engineering & Reproducibility; §9 Conclusion. All numbers use the verified facts (full-precision JSON floats in Table 3 and the chart).

Two adaptations I made:

1. Registered ``C:/Windows/Fonts/arial.ttf`` (+ bold/italic, with a ``registerFontFamily`` mapping) so `??/?±/×/??` render as real glyphs, not black boxes ? confirmed visually across pages 1, 2, 4, 5.
2. Used ``BaseDocTemplate`` with two `PageTemplates` and a ``NextPageTemplate("Later")`` so page 1 gets a clean navy banner (with a white in-banner caption) while pages 2-7 get the running top header + footer page numbers; pushed the title block down 3 cm to clear the banner.